

Uncertainty Sets for Robust Process Rewards

Updated Lit Review, Foundations, and Approaches

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Abstract

We would like to investigate whether an uncertainty set around a learned process reward can reduce reward hacking under inference-time search or reinforcement learning. The challenge is to cover relevant reward errors after optimization while retaining useful discrimination between reasoning paths. These notes distinguish established findings from proposed methods, identify close prior work, explain the necessary foundations, and specify possible empirical and theoretical approaches.

Contents

I	Literature Review	4
1	Most recent impactful developments	4
1.1	Research question and shared notation	4
1.1.1	Shared notation	4
1.2	Uncertainty modeling and pessimistic reward optimization	4
1.3	Reward hacking diagnostics and adversarial defenses	6
1.4	New supervision targets and reasoning-state models	8
1.5	Robust objectives and calibration theory	10
2	Revised literature	11
2.1	Closest predecessors and the possible contribution	12
2.2	Process supervision: targets, data, and representation	12
2.3	Credit assignment and adversarial failure discovery	14
2.4	Reward hacking theory and optimization pressure	15
2.5	Algorithmic context for the experiments	16
2.6	Uncertainty and pessimism: established predecessors	17
2.7	Statistical foundations for constructing and calibrating a tube	19
2.8	Robust optimization and reward shaping	23
2.9	Benchmarks and what their labels can establish	24
2.10	Synthesis and research positioning	25
2.10.1	A defensible candidate contribution	26

II	Notes: Foundations for the Project	27
3	Notation and the objectives we must keep separate	27
4	Reinforcement learning in the reasoning setting	27
4.1	MDPs, states, actions, and policies	27
4.2	Returns, value, and credit assignment	28
4.3	Policy gradients, PPO, and GRPO	28
4.4	RLHF, RLVR, and inference-time search	29
5	What a process reward model does	29
5.1	Labels and training	29
5.2	Aggregation creates additional incentives	30
6	Why reward hacking occurs and why it matters	30
6.1	A numerical example	30
6.2	Different mechanisms need different diagnostics	30
7	Uncertainty, calibration, and coverage	31
7.1	Three kinds of uncertainty	31
7.2	Prediction intervals are not confidence bands for a function	31
8	Robust optimization: the geometry matters	31
9	Distribution movement and reward shaping	32
10	Research habits that make the results interpretable	33
III	Constructing and Characterizing the Reward Tube	34
11	From the line metaphor to a research object	34
11.1	The design specification	34
12	Approach A: problem-level calibration of candidate tubes	35
12.1	Construction	35
12.2	The target problem, and how to handle it	35
12.3	Using the tube	36
12.4	Tests, limitations, and novelty	36
13	Approach B: a coupled function tube from frozen features	36
13.1	A tractable model and an exact induced tube	37
13.2	Joint geometry and robust scoring	37
13.3	Misspecification and practical construction	38
13.4	Research payoff and risks	38
14	Approach C: structured nuisance and budgeted error tubes	38

15 Approach D: a joint reward distribution and robust risk	39
16 Approach E: optimize, audit, and update the tube	40
17 Theoretical characterization: established lemmas and open targets	41
17.1 Why marginal coverage alone fails	41
17.2 A shift-aware bound with transparent assumptions	42
17.3 Open theoretical targets	42
18 An experiment plan that can support a paper	42
18.1 Hypotheses and staged scope	42
18.2 Data and split protocol	43
18.3 Baselines and ablations	43
18.4 Optimization stress tests	44
18.5 Metrics and statistical reporting	44
18.6 Synthetic experiments with known ground truth	45
19 Recommended route and publication decision gates	45
References	47

Part I

Literature Review

1. Most recent impactful developments

This section prioritizes recent work with direct implications for our reward-tube project. The ordering reflects relevance to our research question rather than a bibliometric ranking; recent preprints are distinguished from conference publications in their entries.

1.1. Research question and shared notation

Question. Can a calibrated, structured uncertainty set around a process reward model (PRM) reduce reward hacking under inference-time search or reinforcement learning, while retaining useful discrimination between reasoning paths? We compare with existing pessimistic methods at matched compute, annotation budget, aggregation, and policy movement.

This review was expanded on October 7, 2026. Each entry gives the mathematical object being modeled, important conditions or empirical limitations, and a concrete connection to our project. Equations use consistent notation rather than always reproducing the paper’s symbols. Source section/equation pointers refer to the specified version where given; numbering may change in later revisions. A *schematic* objective describes a method’s structure rather than an exact implementation. Statements introduced as *our interpretation*, *our diagnostic*, or *our extension* are research synthesis, not the authors’ results. We inspected relevant method and theory sections; this is not a complete proof audit or a reproduction of the empirical findings.

1.1.1 Shared notation

Let x be a problem, a_t a reasoning step, $h_t = (x, a_{1:t})$ its prefix, and $\tau = (a_1, \dots, a_T)$ a trajectory. Write $\hat{r}(h_t)$ for a learned process score and $G(\hat{r}(\tau))$ for its trajectory aggregation. Intrinsic step validity, validity of the maintained reasoning state, and

$$p_{\pi_c}(h_t) = \mathbb{P}_{\pi_c}\{\text{correct final answer} \mid h_t\}$$

are distinct targets; π_c is a specified continuation policy. For additive rewards, $J_r(\pi)$ denotes expected return. When using normalized discounted occupancies, d_π is a probability distribution and $J_r(\pi) = \mathbb{E}_{d_\pi} r$. Set $\Delta J_r = J_r(\pi) - J_r(\pi_0)$ for a reference policy π_0 . Two recurring losses are

$$\ell_{\text{BCE}}(y, p) = -y \log p - (1 - y) \log(1 - p), \quad \rho_\beta(u) = u(\beta - \mathbf{1}\{u < 0\}).$$

The second is the pinball loss for a β -quantile. A prediction interval covers a future observed label; a function confidence set aims to contain one unknown reward function. Neither term should substitute for the other.

1.2. Uncertainty modeling and pessimistic reward optimization

Ma et al., Distributional PRMs via Conditional Optimal Transport (2026; arXiv v1).

Model. PRM features condition a one-dimensional OT map. Convex potentials parameterized by

partially input-convex networks give

$$T_\theta(u, h) = \partial_u g_\theta(u, h), \quad T_\theta(\cdot, h)_{\#} \mu_h \approx \nu_h.$$

Here ν_h models rollout-success targets. The conditional dual optimization trains the potentials; a resulting quantile function $Q_\theta(\beta | h)$ can be queried at arbitrary levels [25].

Mathematical detail. Convexity makes the transport map nondecreasing, avoiding quantile crossing. For a general continuous source CDF F_{μ_h} , standard OT quantile extraction is $Q_{\nu_h}(\beta) = T(F_{\mu_h}^{-1}(\beta), h)$; $T(\beta, h)$ itself is a quantile only with the corresponding uniform-source parameterization. Monotonicity is a structural property, not a finite-sample coverage theorem.

Evidence and connection. The paper reports calibration and adaptive budget improvements using empirical completion-success labels. Keep its calibration/evaluation split explicit. Our extension needs joint process uncertainty, calibrated model discrepancy, or post-selection assurance; learning a flexible distribution or monotone quantiles is already this paper’s contribution. Changing the completer changes its target.

Yu et al., From Curiosity to Caution (ICLR 2026; arXiv v1). Score and learning. Random-network distillation fits predictor P_θ to fixed target features T on reference responses. At inference,

$$\alpha(x, y) = \|P_\theta(x, y) - T(x, y)\|^2, \quad r_{\text{caution}} = \hat{r} - \lambda\alpha, \quad \hat{i} = \arg \max_{i \leq N} r_{\text{caution}}(x, y_i).$$

Section 2.3 gives a stylized linear-reward/RND analysis whose informal theorem states a reward gain of order $\sqrt{\log N}$ over ordinary BoN under the stated sampling and network conditions [26].

Conditions and evidence. The theory is a proof of concept under strong distribution/feature assumptions, not a theorem for arbitrary LLMs. The penalty measures reference novelty: unusual valid reasoning may be penalized, and familiar invalid reasoning can evade it. Predictor training needs unlabelled reference responses; scoring requires extra network passes.

Connection. An important low-cost uncertainty comparator. Check whether calibrated residual uncertainty improves on novelty alone, at matched generation and scoring cost. The name “LCB” does not establish calibrated reward-error coverage in the actual PRM setting.

Hahami et al., A Unifying Lens on Reward Uncertainty (2026; arXiv v1). Variational formula. For believed reward distribution P , a KL-penalized reward adversary gives

$$r_{\text{eff}} = \inf_Q \{\mathbb{E}_Q r + \eta D_{\text{KL}}(Q \| P)\} = -\eta \log \mathbb{E}_P e^{-r/\eta}.$$

The optimizer exponentially tilts P toward lower rewards. For a uniform ensemble, $r_{\text{eff}} = -\eta \log[K^{-1} \sum_k e^{-r_k/\eta}]$; see Section 3 [27].

Limits and assumptions. Under suitable moments, $r_{\text{eff}} = \mathbb{E}r - \text{Var}(r)/(2\eta) + O(\eta^{-2})$ as $\eta \rightarrow \infty$; $\eta \downarrow 0$ gives the essential minimum. The Gaussian expression is exact when r really is Gaussian. Exponential moments must exist. Reward adversary temperature η and policy KL coefficient have different roles, even if a particular formulation sets them equal. A hard-radius KL ball instead requires optimizing a dual multiplier.

Connection. Mean, minimum, and variance penalties already share an entropic formulation. Our extension should model a *joint* step-reward law and its calibration, rather than rediscover scalar log-sum-exp pessimism. An incorrect believed distribution remains incorrect after robust tilting.

Zhou et al., PRISM (ACL 2026). **Model and likelihood.** Frozen features feed Gaussian expert heads and a prompt-dependent router:

$$p(r \mid x, y) = \sum_{k=1}^K \omega_k(x) \mathcal{N}(r; \mu_k(x, y), \sigma_k^2(x, y)).$$

Stage 1 forms experts; Stage 2 freezes them and learns router weights. Its cross-expert preference likelihood averages $\Phi((\mu_k(y_+) - \mu_l(y_-))/\sqrt{\sigma_k^2(y_+) + \sigma_l^2(y_-) + \epsilon})$ with weights $\omega_k \omega_l$, where Φ is the normal CDF; Section 3, Eqs. (8)–(10) [28].

Conditions and interpretation. Mixture structure represents preference heterogeneity and within-component dispersion. These components are not identified epistemic versus aleatoric uncertainty solely by their names; latent mixtures and approximate likelihoods need empirical validation. Gaussian reward differences use the model’s independence convention.

Connection. A rich-distribution comparator for preference rewards, not a certified process-verification method. Our extension could share latent expert weights across steps, creating coherent trajectory-level scenarios. Decide whether uncertainty concerns conflicting legitimate preferences or incorrect mathematical judgments; averaging them can obscure the target of the tube.

Duan et al., Bayesian Non-negative Reward Modeling / BNRM (2026; arXiv v1). **Generative model.** Local activations $z \geq 0$ and a shared dictionary $\phi \geq 0$ have Gamma priors, generate reward $r = z^\top \phi$, and define preference probability $\text{sigmoid}(r_+ - r_-)$. Weibull variational posteriors are fit using the regularized evidence objective

$$\mathbb{E}_q \log p(\mathcal{D} \mid z, \phi) - \eta D_{\text{KL}}(q(z) \parallel p(z)) - \eta D_{\text{KL}}(q(\phi) \parallel p(\phi)).$$

See Section 4, Eqs. (5)–(9) [29].

Conditions and evidence. Nonnegative sparse factors encourage parts-based representations; they do not prove causal disentanglement or elimination of style bias. A variational posterior can underestimate uncertainty, and misspecification can persist across all factors. Preference rewards are identified through differences, so positivity alone is not a correctness certificate. Reported robustness gains concern the evaluated RLHF settings.

Connection. A possible alternative to Gaussian heads or independent ensembles. A shared dictionary creates a natural source of correlated error across prefixes; local latent randomness and global uncertainty should be separated when constructing trajectory scenarios. Any posterior credible tube still needs independent coverage and optimization stress tests.

1.3. Reward hacking diagnostics and adversarial defenses

Tiwari et al., Reward Under Attack (2026; arXiv v1). **Setup and attack objective.** The paper uses model-specific aggregation: Skywork’s terminal score and Qwen’s minimum step score.

Adversarial probing seeks high-scoring token modifications; in our notation its attack target is

$$\max_{u \in \mathcal{A}} G(\hat{r}(\tau \oplus u)),$$

subject to the attack’s permitted placements and token budget. Static perturbations, gradient-based token optimization, and GRPO expose different failure mechanisms [12].

Evidence and limits. In the studied Skywork/AIME setting, reward approaches one while answer accuracy remains below 4%. Qwen resists the particular gradient attack but collapses during RL to a vacuous introductory sentence. Therefore success against one attack is not general robustness. PRM-BiasBench tests controlled content and presentation changes.

Connection. Optimize the *deployed robust score*, not only the nominal PRM. Our diagnostic should report reward, independent correctness, claim count, length, and vacuity across pressure levels. Parser exploits and reward inflation require separate tests; a score tube that covers only mathematical steps may miss an output that avoids making any claim.

Juneja et al., Adversarial Training for PRMs / APRM (ICML 2026). Game and objective. A generator corrupts a reference step; a verifier predicts correctness $\hat{y}$, checked by an algorithmic oracle y . Verifier payoff is $r_R = 21\{y = \hat{y}\} - 1$. Generator payoff is +1 for an incorrect accepted step, 0 for an incorrect rejected step, and -1 for an uncorrupted correct step. Each player optimizes

$$U'_i = \mathbb{E}r_i - \tau D_{\text{KL}}(\pi_i \| \pi_{i,0}) + c_H H(\pi_i).$$

The inspected preprint v1, Section 2, formulates a general-sum game [13].

Theoretical conditions. Finite action spaces, continuity, and concavity conditions support equilibrium arguments in mixed-policy space. Own-player strong concavity is not by itself enough to make a general-sum pseudo-gradient strongly monotone; cross-player derivatives matter. Neural PPO optimization does not automatically inherit those policy-space properties. The oracle uses reference-based semantic/structural checks and has its own error modes.

Connection. A strong hard-negative baseline and adversarial data source. Compare the same annotation budget spent on retraining versus tube construction; audit oracle mistakes before treating generated corruption labels as truth.

Pathmanathan and Huang, REFORM (ACL 2026; inspected arXiv v1). Method and objective. REFORM discovers preference-order reversals while attempting to preserve response quality. An aligned policy proposes top- k tokens and reward-guided decoding searches for low-scoring preferred responses; a misaligned policy searches for high-scoring nonpreferred responses. Failure-aware retraining minimizes Bradley–Terry loss,

$$\mathcal{L}_{\text{BT}} = -\mathbb{E} \log \text{sigmoid}(r_\phi(x, y_+) - r_\phi(x, y_-)),$$

on original and discovered preference pairs [14].

Conditions and evidence. Section 3 substitutes a prefix reward for an intractable expected completion reward during controlled decoding. Policy likelihood and known preference classes constrain the search, but high likelihood does not certify class preservation. New examples still need reliable preference labels. The v1 title differs from the final bibliography title.

Connection. Optimizer-discovered failures are useful calibration or misspecification examples. Our extension should compare against retraining on those same examples. Maintain an independent audit split: fitting both the reward and its tube to discovered attacks can overfit the tested attack family.

1.4. New supervision targets and reasoning-state models

Xie et al., Noise-aware Learning (2026; arXiv v1). **Method and formula.** Reflection-aware correction rejects a successful continuation as evidence for step t if a later step explicitly repairs t . Noise-aware iterative training then updates a filtered label via

$$\tilde{y}_t^{(k+1)} = \begin{cases} r_t^{(k)}, & |r_t^{(k)} - \tilde{y}_t^{(k)}| > \delta, \\ \tilde{y}_t^{(k)}, & \text{otherwise,} \end{cases}$$

where $r_t^{(k)}$ is the current PRM prediction. This is the refinement rule in Eq. (7), followed by retraining [4].

Conditions and evidence. The correction targets MC false positives caused by repair; limited completer capability can still produce false negatives. Selective self-relabeling assumes that, after filtering, large disagreement is more often label noise than model error. If that assumption fails, the procedure can reinforce confident mistakes. Reported step-discrimination gains are empirical, not finite-sample guarantees.

Connection. Treat reflection indicators and label/model disagreement as candidate uncertainty features. Evaluate against independent human or formal labels; the model’s own refined labels cannot also serve as the truth used to validate its tube.

Wang et al., Hierarchical Multi-Step Reward Models (arXiv v5, 2026). **Model.** HRM augments single-step supervision with consecutive merged steps. Its basic dataset construction is

$$\mathcal{D}_{\text{HRM}} = \mathcal{D}_{\text{PRM}} \cup \{(a_i \oplus a_{i+1}, R(a_i \oplus a_{i+1}))\}_{i < T},$$

where $\oplus$ concatenates steps. Hierarchical node compression augments MCTS-derived supervision. A two-class softmax produces process scores [49].

Conditions and evidence. Section 3.2’s compression argument preserves a sample-mean expectation under its i.i.d.-child assumption, while changing variance from σ^2/N to $\sigma^2/(N-1)$ when one child is removed. Correlated search children need separate treatment. The policy-improvement experiments use filtered-data supervised fine-tuning rather than PPO/GRPO; stronger search performance is not an RL robustness theorem.

Connection. Merged-step supervision is a representation baseline for inter-step dependence. Our structured tube should improve on a well-trained multi-step center, not merely on an impoverished independent-step model. Ordinary prefix-conditioned PRMs already see history; explicit multi-step supervision changes the training signal, not the existence of prefix context.

Han et al., Cliff (September 2026; arXiv v1). **Method and formula.** A teacher solves and verifies a problem, then identifies the first pitfall in incorrect student rollouts. With binary group

success mean μ and $\sigma = \sqrt{\mu(1-\mu)}$, define $A_{\text{cor}} = (1-\mu)/\sigma$ and $A_{\text{inc}} = -\mu/\sigma$. Cliff assigns

$$A_{ij} = \begin{cases} A_{\text{cor}} - b, & \text{correct rollout,} \\ \lambda A_{\text{cor}} - b, & \text{incorrect, before pitfall,} \\ A_{\text{inc}} - b, & \text{incorrect, at/after pitfall.} \end{cases}$$

Here b is its centering baseline and λ adjusts prefix credit; see Section 3.2, Eqs. (3)–(4) [7].

Conditions and evidence. Incorrect teacher solutions trigger a vanilla-GRPO fallback. Homogeneous outcome groups need the implementation’s zero-variance convention. Teacher validity and error localization are substantive assumptions; a correct teacher answer does not certify every judgment.

Connection. Put uncertainty on the pitfall index rather than only scalar step scores. Compare prefix/suffix credit with robust scoring. The first-error boundary assumes a useful decomposition; repairing an earlier mistake can make that decomposition too coarse.

Gan et al., Reasoning State Propagation (September 2026; arXiv v1). Model and formula. RSP predicts break probability b_t and repair probability u_t for valid/invalid maintained reasoning states. With row-vector state probabilities,

$$M_t = \begin{pmatrix} 1 - b_t & b_t \\ u_t & 1 - u_t \end{pmatrix}, \quad \mathbf{p}_t = \mathbf{p}_{t-1} M_t, \quad \mathbf{p}_0 = (1, 0).$$

The process score is p_t^{valid} . BCE on annotated intermediate states and on the terminal outcome trains the propagated probabilities; Section 3, Eqs. (2)–(9) [8].

Conditions and evidence. Outcome gradients reach earlier transitions through the matrix product. Process labels ground intermediate states; outcome-only data alone generally cannot identify every break/repair probability. Equating a terminal valid state with answer correctness is a modeling decision that can fail for lucky answers or incomplete reasoning.

Connection. A promising structured tube is a set of jointly admissible transition matrices, not independent score intervals. Our extension could study how transition uncertainty propagates and how ignoring repair causes excessive pessimism. This requires a separate comparison with RSP and labels that distinguish local errors from unresolved errors.

Fan et al., TIPS (September 2026; arXiv v1). Method and theoretical detail. A generative verifier emits thought Z , step labels $\hat{\mathbf{Y}}$, and outcome judgment $\hat{D}$. GRPO uses only $r = \mathbf{1}\{\hat{D} = D\}$ to update the whole output. With pre-thought representation H , outcome error P_e , and binary entropy H_b , the paper’s combined information bound is

$$I(Z; \mathbf{Y} \mid H) \geq H(D \mid H) - H_b(P_e) - H(D \mid \mathbf{Y}, H).$$

This follows from its Theorems 1–2, Section 3.3 [9].

Conditions and evidence. The bound is informative when verification is nontrivial, outcome error is small, and process labels largely determine the outcome. It concerns information in the thought, not correctness of each emitted label, calibrated probabilities, or resistance to reward optimization. Thought may encode the relevant information without the parser exposing it reliably.

Connection. A potentially inexpensive center or independent critic. Audit generated labels separately and measure stochastic-judgment variability. Our uncertainty set cannot inherit process validity from outcome supervision without these additional checks.

1.5. Robust objectives and calibration theory

Liu, Sun, and Zheng, Robust Optimization with Correlated Proxies (ICLR 2026).

Uncertainty set and exact objective. Normalize proxy z to mean zero and variance one under d_{π_0} . The paper constrains plausible rewards to reference mean M , standard deviation $V > 0$, and correlation ρ with z . Let $a = \mathbb{E}_{d_\pi} z$ and $D = \chi^2(d_\pi \| d_{\pi_0})$. Its inner solution gives

$$\inf_{r \in \mathcal{U}_{\text{corr}}} J_r(\pi) = M + V \left[\rho a - \sqrt{1 - \rho^2} \sqrt{D - a^2} \right].$$

See the proceedings paper, Section 3.1, Eqs. (4)–(9) [19].

Conditions and theory. Support overlap and reference moments are essential; outside reference support, the unconstrained adversary can make rewards arbitrarily poor. The subtraction $D - a^2$ removes movement along the normalized proxy direction. A known-feature linear variant restricts reward structure and improves interpretability. The paper also analyzes occupancy estimation and optimization.

Connection. A directly relevant example of *coupling*, not an independent interval construction. Our extension would estimate a process-informed set from supervision and compare its geometry with this correlation set. Generic max–min reward optimization and safe-reference improvement are already prior work.

Liu et al., Robust General Utility for RL (August 2026; arXiv v1). **Objective.** The paper uses costs $f_\xi(\lambda_\theta)$ of policy occupancy λ_θ and solves

$$\min_{\theta \in \Theta} \max_{\xi \in \Xi} f_\xi(\lambda_\theta), \quad \Xi = \{\xi : d(\xi, \hat{\xi}) \leq d_0\}.$$

Reward maximization reverses these signs. This includes linear rewards but also nonlinear occupancy utilities; see Section 3, Eq. (2) [20].

Theoretical conditions. The analysis separates utility regimes and uses convex compact parameter sets, smoothness conditions, and appropriate stationarity measures; proximal methods handle more general minimax cases. A stationarity guarantee is not a global-optimality or true-reward guarantee for arbitrary neural policies. The actual evaluation utility must belong to the modeled ambiguity set for the robust interpretation to hold.

Connection. Useful when uncertainty concerns a utility beyond additive step reward. However, trajectory minimum/product aggregation is not automatically a function of ordinary state-action occupancy: history augmentation or a richer trajectory distribution may be required. Our formulation should establish that representation before importing general-utility algorithms or convergence claims.

Conformal prediction through hypothesis testing [38]. Theoretical viewpoint. Invert a family of level- α tests of a candidate augmented observation:

$$C_n(x) = \{y : \varphi(Z_{1:n}, (x, y)) = 0\}.$$

If the test rejects the true augmented observation with probability at most α , then $\mathbb{P}\{Y_{n+1} \in C_n(X_{n+1})\} \geq 1 - \alpha$. The paper studies universality of permutation-based constructions under exchangeability, impossibility of unrestricted conditional coverage, and oracle optimality. Its oracle construction uses inverse conditional-density scores.

Conditions and limits. Optimality is relative to a specified distribution and efficiency criterion; the conditional density is ordinarily unknown. Marginal exchangeability-based validity does not imply useful coverage at every rare input. Continuous interval-size objectives also need adaptation for binary labels and finite prediction sets.

Connection. A useful conceptual check on proposed tube guarantees: state the null being tested, the sampling symmetry, and the coverage event. A valid test for a future random outcome does not by itself yield a simultaneous band for the conditional mean reward. The impossibility results motivate restricted subgroups, explicit structural assumptions, or approximate conditional guarantees.

CQR and minimax limits under known covariate shift [39]. Theoretical detail. The paper analyzes conditional coverage and length errors relative to oracle equal-tailed intervals, separating quantile-estimation, calibration, and rank-discretization effects. It assumes known bounded density ratios in its shift results. In its explicit fixed-score, single-carrier construction, the minimax calibration scale is

$$\mathcal{R}_{m,p} \asymp \sqrt{\frac{1 + \chi^2(Q_X \| P_X)}{m}}.$$

Its multi-atom construction adds a complexity factor corresponding to the number of atoms. This suggests the information scale $m_{\text{eff}} = m / (1 + \chi^2(Q_X \| P_X))$.

Conditions and limits. These are results for the paper’s specified classes and fixed-score calibration setting. They are not a universal lower bound for every learned representation, reward estimator, or adaptive PRM procedure. A marginal guarantee alone does not bound the realized conditional coverage profile.

Connection. Helps quantify why highly concentrated optimized policies can demand substantially more calibration data. A theoretical project could specialize these error decompositions to problem-level trajectory scores or selected-prefix populations, while explicitly checking bounded-ratio and score assumptions. The divergence-based effective sample size is a diagnostic, not an independent proof of trajectory coverage.

2. Revised literature

The following literature supplies the established methods, theoretical foundations, and evaluation resources needed to interpret the recent developments. Entries retain their mathematical details and limitations; papers already discussed above are not repeated.

2.1. Closest predecessors and the possible contribution

Uncertain rewards and pessimistic optimization are established ideas. Four comparisons are especially important:

1. AdvPO constructs a shared last-layer confidence set and optimizes reference-centered pessimistic improvement [22].
2. Quantile and conditional-OT PRMs estimate prefix continuation-success uncertainty [24, 25].
3. Ensembles and reward-distribution DRO yield conservative effective rewards [21, 23, 27].
4. JOMI addresses prediction coverage after data-dependent selection [33]; correlation-constrained robust RL already supplies structured reward sets and safe-improvement bounds [18, 19].

Our proposed contribution must therefore concern *process-specific targets, shared error geometry, coverage after optimization, or an informative failure characterization*, rather than the generic idea of placing a tube around a reward.

2.2. Process supervision: targets, data, and representation

Lightman et al., Let’s Verify Step by Step (2023). Model and objective. Human annotators label steps as positive, negative, or neutral. The PRM predicts label probabilities conditioned on the problem and prefix; training is step-label cross-entropy. The preferred solution score in Appendix F is

$$S(\tau) = \prod_{t=1}^T p_{\theta}(\text{positive or neutral} \mid h_t).$$

The paper compares this with minimum aggregation and different handling of neutral labels, and evaluates best-of- N selection on mathematical solutions [1].

Conditions and evidence. PRM800K supplies human process supervision; active learning concentrates annotation on informative candidates. Annotation protocols and termination at a first error affect which prefixes receive labels. The product is a ranking rule: it is a joint-validity probability only if its factors have the appropriate conditional-event interpretation. It also penalizes longer trajectories.

Connection. Use this as the intrinsic-validity data anchor and nominal PRM baseline. Our tube must define its treatment of neutral labels and censored post-error steps. Strong candidate-selection performance does not establish coverage or robustness under an RL-trained optimizer.

Wang et al., Math-Shepherd (ACL 2024; arXiv v2). Model and objective. A completer samples K continuations from each prefix. With C_k indicating a correct final answer, the two annotation rules are

$$\hat{y}_t^{\text{SE}} = K^{-1} \sum_k C_k, \quad \hat{y}_t^{\text{HE}} = \mathbf{1}\{\max_k C_k = 1\}.$$

The verifier learns these labels and supports reranking and process-supervised PPO; see the annotation section and Eqs. (3)–(4) [2].

Conditions and evidence. Under independent Bernoulli continuations, soft estimation has mean $p_{\pi_c}(h_t)$ and variance $p_{\pi_c}(1 - p_{\pi_c})/K$. Our calculation gives $\mathbb{E}\hat{y}_t^{\text{HE}} = 1 - (1 - p_{\pi_c}(h_t))^K$. Thus

hard labels depend on both completer capability and rollout count; they are not unbiased estimates of the single-rollout success probability. A repaired incorrect step can receive a positive label.

Connection. Separate finite-rollout noise from incorrect target choice. Increasing K shrinks estimation noise for the soft target but does not make completion success equal logical correctness. Freeze the completer when calibrating a success-probability tube.

Zhang et al., The Lessons of Developing PRMs (2025; arXiv v1). **Method.** The study compares Monte Carlo labels, LLM judgments, and human annotations. Consensus filtering retains solutions when MC and the judge agree on the first erroneous step. In consistent notation, its selection condition is

$$\mathcal{D}_{\text{keep}} = \{\tau : k_{\text{MC}}(\tau) = k_{\text{judge}}(\tau)\}.$$

The trained PRMs use step classification; the study also compares soft MC labels with hard labels $y_t = \mathbf{1}\{\tilde{p}_t > 0\}$ and evaluates both best-of- N and ProcessBench [3].

Conditions and evidence. In the reported eight-completion experiments, hard labels perform particularly well after filtering. Agreement is a quality filter, not proof that both annotators are correct. Conditioning on agreement also changes the data distribution and can discard difficult cases. Final-answer ranking and first-error identification need not improve together.

Connection. Keep corrected supervision identical across uncertainty methods. Audit the rejected and retained strata: a tube calibrated only on consensus examples may fail on precisely the disputed prefixes that an optimizer finds. This paper supplies practical Qwen PRM comparators, not a confidence-set guarantee.

Ding et al., SCAN (2025; arXiv v1). **Method and formula.** SCAN estimates problem-level self-confidence $SC_\pi(x)$ as the fraction of correct sampled answers, uses high-confidence positive examples, and concentrates stepwise MC annotation on negatives. Near the estimated first error k_e , its soft-label correction has the form

$$\hat{y}_t = \min\{c_t/SC_\pi(x), 1\}; \quad \hat{y}_t = \mathbf{1}\{c_t > 0\} \text{ away from that region,}$$

where c_t is continuation success. A tolerance distance controls the affected steps; see Section 3.2, Eqs. (5)–(7) [5].

Conditions and evidence. Confidence-based filtering reduces annotation cost, while soft labels near a boundary accommodate error-location noise. Division by self-confidence is a heuristic capability correction, not an identified importance ratio or an unbiased estimator of intrinsic validity. Its denominator must be nonzero; filtering is part of the protocol.

Connection. A useful comparator for improving the reward center and annotation efficiency. Stratify tube errors by problem confidence and distance to the estimated error. Check difficult low-confidence problems independently, since training selection can hide their uncertainty.

Pala et al., Error Typing for Smarter Rewards / PathFinder-PRM (2025; arXiv v1).

Model and objective. PathFinder decouples error detection from reward estimation using two passes:

$$(M_t, C_t) = f_\theta(h_t), \quad R_t = g_\theta(h_t, M_t, C_t).$$

The first predicts mathematical and consistency signals; the second conditions its reward on those signals. Training uses cross-entropy on categorical label tokens for mathematical validity, consistency, and step correctness [6].

Conditions and evidence. Section 3.2 calls the dataset approximately 400K trajectory samples, assembled from PRM800K and Mistral-PRM data; the precise serialized sample unit matters for annotation-cost comparisons. The paper maps neutral PRM800K labels into a category interpreted as suboptimal, which is its labeling convention rather than a universal definition of neutral validity. Both passes may share representation and judgment errors.

Connection. Typed residuals can define distinct tube directions or coverage strata. Test whether any gain comes from the uncertainty geometry or simply richer labels. Account for the second forward pass when matching compute; error-type predictions do not themselves certify reward bounds.

2.3. Credit assignment and adversarial failure discovery

Cheng et al., Stop Summation / PURE (NeurIPS 2025; arXiv v1). Objective and implementation. PURE targets $\max_{\pi} \mathbb{E}_{\pi}[\min_t r_t]$, rather than a sum. Its transformed step reward is

$$\omega_t = \frac{\exp(-r_t/\vartheta)}{\sum_j \exp(-r_j/\vartheta)}, \quad r_t^{\text{tr}} = \omega_t r_t.$$

For a unique worst step w , $\vartheta \downarrow 0$ concentrates reward at w . With $\gamma = 1$, return-to-go is r_w before/at w and zero afterwards. Section 3.2 adds a leave-one-out baseline and optionally verifiable outcome reward [10].

Conditions and evidence. The exact minimum correspondence is a zero-temperature limit; finite temperature is a smooth reweighting. Ties share weight across worst steps. Minimum aggregation limits accumulation of favorable errors but can reward vacuous outputs with no low-scoring mathematical claims. It also treats a weak step differently from a recoverable reasoning error.

Connection. Required aggregation/credit-assignment baseline. Hold G fixed before comparing uncertainty estimators. Our extension could apply minimum aggregation to lower bounds, but any improvement must exceed simply switching from sum to minimum or adding an outcome check.

Zheng et al., CoLD (arXiv v2, May 2026). Model and objective. A noisy-feature bias estimator gives $q_{\phi}(h) = \text{sigmoid}(b_{\phi}(h + \mathcal{N}))$. The inference score in Eq. (3) is

$$r_{\text{deb}}(h) = r_{\theta}(h)q_{\phi}(h) - cq_{\phi}(h) - \alpha\ell(h),$$

where ℓ measures step length. Joint training uses correctness cross-entropy and complementary correlation terms $\lambda_r \text{Corr}(r_{\theta}, \ell)^2$ and $-\lambda_b \text{Corr}(b_{\phi}, \ell)^2$ [11].

Conditions and evidence. The estimator is intended to capture length and surface style; Gaussian feature noise approximates the counterfactual removal of semantics. That approximation does not identify a causal length effect without further assumptions. The paper’s limitation discussion recognizes this issue. A deterministic length penalty also changes preferences between long and short correct solutions.

Connection. Length is a concrete common error direction for a structured tube. Compare against this direct bias correction and measure residual errors under meaning-preserving verbosity changes. A learned bias parameter’s uncertainty would need independent estimation; neither the correlation penalty nor injected noise supplies confidence coverage.

Wallace et al., Universal Adversarial Triggers (EMNLP 2019). Attack construction.

A short shared token sequence is optimized to cause a target behavior across many inputs. The objective can be written $\min_u \mathbb{E}_{x \sim \mathcal{D}} \mathcal{L}(f(u \oplus x), y_{\text{target}})$. A first-order token replacement ranks vocabulary embeddings using

$$v \in \arg \min_{v' \in \mathcal{V}} (e_{v'} - e_{u_j})^\top \nabla_{e_{u_j}} \mathcal{L},$$

then evaluates candidate replacements and iterates [15].

Conditions and evidence. This is a gradient-based discrete search approximation, not a proof that the chosen trigger globally optimizes the loss. The original demonstrations concern classification, reading comprehension, and language generation; transfer to a particular PRM is an empirical question. A universal trigger differs from optimizing a separate attack for each problem.

Connection. Test shared prefixes, suffixes, and reward-delimiter perturbations in addition to semantic mistakes. Shared attacks can reveal a common model bias that ensemble spread misses. Keep the permitted token locations explicit; formatting attacks may exploit the scorer’s parser rather than uncertainty about the mathematical reasoning itself.

2.4. Reward hacking theory and optimization pressure

Skalse et al., Defining and Characterizing Reward Hacking (arXiv v2, 2025). Definition.

Proxy $\hat{r}$ and target r^* are hackable over policy class Π if there exist $\pi, \pi' \in \Pi$ with

$$J_{\hat{r}}(\pi') > J_{\hat{r}}(\pi), \quad J_{r^*}(\pi') < J_{r^*}(\pi).$$

Additive returns are linear in discounted visit counts: $J_r(\pi) = \langle r, \mathcal{F}^\pi \rangle$ [16].

Theoretical result. The revised full text’s Theorem 1 states that nontrivial unhackable reward pairs are equivalent on a stationary-policy set containing an open subset; Corollary 1 covers all stationary policies. Equivalent means the same policy ordering, not identical numerical rewards. Finite policy sets permit nontrivial nonequivalent unhackable pairs. The abstract’s compressed wording should not replace these precise statements.

Connection. Aim for restricted-domain coverage or safe improvement relative to a reference, rather than universal nontrivial proxy alignment. A finite candidate pool has a different guarantee scope from all policies. Our tube can reduce specific reversals without eliminating every possible reversal or implying that every nominal RL run must hack.

Gao et al., Scaling Laws for Reward Model Overoptimization (2022). Empirical formulas. With $d = \sqrt{D_{\text{KL}}(\pi || \pi_0)}$, gold-model reward improvement is fitted by

$$R_{\text{BoN}}(d) = d(\alpha_{\text{B}} - \beta_{\text{B}}d), \quad R_{\text{RL}}(d) = d(\alpha_{\text{R}} - \beta_{\text{R}} \log d).$$

Coefficients vary with reward-model size and data. For continuous, untied best-of- N selection, the paper uses $D_{\text{KL}}^{\text{BoN}} = \log N - (N - 1)/N$ [17].

Conditions and evidence. These are fitted scaling relationships, not universal laws. The RL expression is not a good local description at the origin. A larger learned gold model supplies synthetic preferences and evaluation; it is still a proxy for human intent. The analytic BoN KL formula assumes the particular independent-sampling/ranking model and needs care with ties or restricted discrete supports.

Connection. Plot target quality against N , training steps, and policy divergence, alongside proxy reward. Compare the point at which quality peaks and declines. Mathematical answer checks and independent process audits can replace a learned gold reward where available; one favorable checkpoint cannot establish mitigation.

Laidlaw et al., Correlated Proxies (ICLR 2025; arXiv v3). Definition and theorem. Let $\text{Corr}_{d_{\pi_0}}(\hat{r}, r^*) = \rho > 0$ and let $\sigma_{\hat{r}}, \sigma_{r^*}$ be reference-distribution standard deviations. For $d_{\pi} \ll d_{\pi_0}$, Theorem 5.1 gives

$$\frac{\Delta J_{r^*}}{\sigma_{r^*}} \geq \frac{1}{\rho} \left[\frac{\Delta J_{\hat{r}}}{\sigma_{\hat{r}}} - \sqrt{(1 - \rho^2) \chi^2(d_{\pi} \| d_{\pi_0})} \right].$$

This motivates occupancy-regularized policy optimization (ORPO) [18].

Conditions. Correlation is under the reference occupancy, not a pointwise error guarantee. Nonzero variances, support overlap, and normalized occupancy conventions matter. A positive lower bound certifies improvement; there need not exist a policy achieving such a bound. Action-level KL and occupancy divergence are different objects.

Connection. Relate tube residual error to the change in prefix occupancy, and test whether policy movement explains apparent mitigation. Estimating a correlation or density ratio in enormous text state spaces is itself a statistical problem. This result supplies a comparison for any proposed reference-centered PRM safe-improvement theorem.

2.5. Algorithmic context for the experiments

Sutton and Barto, Reinforcement Learning: An Introduction (2018). Core equations.

For a fixed policy in a discounted MDP, the Bellman expectation equation is

$$V^{\pi}(s) = \sum_a \pi(a | s) \sum_{s', r} p(s', r | s, a) [r + \gamma V^{\pi}(s')].$$

Temporal-difference learning uses $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ and $V(s_t) \leftarrow V(s_t) + \alpha \delta_t$; the policy-gradient theorem underlies actor-critic learning [46].

Conditions. The state must contain enough history for the Markov property. Infinite-horizon discounted values require appropriate boundedness and $\gamma < 1$; finite-horizon values also depend on time. Standard Bellman recursions assume additive rewards and a fixed model. Approximate values and bootstrapping introduce estimation error separate from PRM error.

Connection. Use Part II to translate these objects into text prefixes, tokens/steps, and terminal correctness. Our tube changes the reward objective, while a critic estimates return under that objective. Confusing critic variance with reward uncertainty would obscure the mechanism. This is a textbook background reference, not a new reward-hacking result.

Schulman et al., Proximal Policy Optimization (2017). **Objective.** With importance ratio $q_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\text{old}}(a_t | s_t)$, PPO’s clipped surrogate is

$$L^{\text{clip}}(\theta) = \hat{\mathbb{E}}_t \min\{q_t \hat{A}_t, \text{clip}(q_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t\}.$$

The full training objective can add value-fitting and entropy terms; the paper also studies adaptive KL penalties [47].

Theoretical distinction. The minimum is a lower surrogate relative to the unclipped sample objective. It is not a confidence bound on true reward and does not enforce a global trust region for every state. Multiple epochs reuse on-policy samples while controlling update incentives; a bad reward still gives a bad advantage estimate.

Connection. Keep clipping, critic fitting, KL coefficients, and optimization epochs fixed when comparing nominal and robust PRM rewards. Recompute the intended robust return/advantage consistently. PPO’s use of “pessimism” concerns an optimization surrogate and must not be conflated with pessimism over an uncertain reward function.

Shao et al., DeepSeekMath / GRPO (2024; arXiv v3). **Objective and credit assignment.** GRPO eliminates a learned value baseline by sampling a group and normalizing its rewards. Outcome advantages are

$$\hat{A}_i = \frac{R_i - \bar{R}}{s_R}; \quad \hat{A}_{i,t}^{\text{process}} = \sum_{j: \text{end}(j) \geq t} \frac{r_{ij} - \bar{r}}{s_r}.$$

These feed a PPO-style clipped objective with a separate policy KL term. The process formula normalizes over group step rewards; see Section 4.1 [48].

Conditions and evidence. Groups with identical outcomes give zero centered advantages and require a zero-standard-deviation convention. Summing future normalized process rewards changes length incentives and credit assignment. Group normalization can also amplify small differences. The paper studies iterative reward-model updates; the target distribution can consequently change during training.

Connection. A natural RL testbed, but a lower PRM score is not by itself a lower bound on the normalized advantage. Audit how our tube’s scores pass through normalization and aggregation. Distinguish reward-model uncertainty, Monte Carlo group variability, and policy-gradient estimation noise.

2.6. Uncertainty and pessimism: established predecessors

Coste et al., Reward Model Ensembles (arXiv v2, 2024). **Aggregation rules.** For K reward models, define $\mu = K^{-1} \sum_k r_k$ and $v = K^{-1} \sum_k (r_k - \mu)^2$. The compared scores are

$$r_{\text{mean}} = \mu, \quad r_{\text{WCO}} = \min_k r_k, \quad r_{\text{UWO}} = \mu - \lambda v.$$

Worst-case optimization (WCO) and uncertainty-weighted optimization (UWO) are tested with best-of- N and PPO; see Section 3, Eqs. (4)–(5) [21].

Evidence and assumptions. Experiments use a larger learned gold reward, including synthetic preference-label flips. Member diversity comes from initialization and training randomness on shared data. Minimum-member scoring is below truth at a point if at least one member is not overestimating

there; the method does not guarantee that condition. Its pointwise minimum also need not correspond to one shared reward model for all trajectories.

Connection. Include all three rules with matched ensemble cost. Distinguish variance from standard-deviation penalties and align reward scales. Test common errors rather than treating member spread as calibrated confidence. Our coupled function set should be compared with both per-sample WCO and a single worst ensemble member chosen at policy level.

Zhang et al., AdvPO (2024; arXiv v2). **Confidence set.** With frozen final features $e(x, y)$, $r_\phi = \phi^\top e$, construct

$$M_D = \lambda I + \sum_{i, y \in \{y_i^+, y_i^-\}} e(x_i, y) e(x_i, y)^\top, \quad \mathcal{C} = \{\phi : \|\phi - \hat{\phi}\|_{M_D} \leq b\}.$$

For $g = \mathbb{E}_\pi e - \mathbb{E}_{\text{ref}} e$, reference-centered robust improvement becomes

$$\inf_{\phi \in \mathcal{C}} \phi^\top g = \hat{\phi}^\top g - b \sqrt{g^\top M_D^{-1} g}.$$

A policy KL penalty completes the objective; Section 4, Eqs. (5)–(7), Theorem 4.1 [22].

Theory and limits. Theorem 3.1’s reward bound includes an approximation term and assumes infinite-width/positive-definite NTK structure. Experimental radii are tuned rather than independently coverage-calibrated. One common parameter adversary is less conservative than separate worst-case choices for each sample, by convexity of the norm; this distinction is already central here.

Connection. The closest predecessor to Approach B and to reference-centered safe improvement. A publishable extension must add process-specific construction, measured joint coverage, or misspecification control, rather than merely repeat the ellipsoid’s dual-norm formula.

Yan et al., Reward-Robust RLHF (2024; arXiv v1). **Objective and uncertainty model.** For a finite reward family $\mathcal{U}$,

$$J_\lambda(\pi) = \lambda J_{\hat{r}}(\pi) + (1 - \lambda) \min_{r \in \mathcal{U}} J_r(\pi).$$

The Bayesian Reward Model Ensemble shares a backbone and has Gaussian heads $r_k = \mu_k + \sigma_k \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 1)$. The nominal head is selected using low uncertainty; see Sections 5–5.1 [23].

Conditions and evidence. The nominal/robust tradeoff reduces excessive conservatism but weakens a purely worst-case interpretation. Shared features can produce common bias, and Gaussian sampling is a modeling assumption, not proof that the true reward lies in the family. The paper’s constant-reward argument requires action-independent remaining return: variable termination or length-dependent cumulative constants need separate treatment. Policy KL can still move the actor even when reward advantages vanish.

Connection. Compare robust tubes with a nominal–robust interpolation, but report the tradeoff parameter and coverage independently. Preserve the minimum-outside-expectation objective when implementing a shared-function adversary; swapping that order changes the method and its conservatism.

Park et al., Know What You Don’t Know (2025; arXiv v1). **Target and calibration.** Fixed-completer rollouts produce $\tilde{p}(h) = K^{-1} \sum_k C_k$. Quantile heads minimize

$$\mathcal{L}_{\text{QR}} = \frac{1}{|\mathcal{B}|} \sum_{\beta \in \mathcal{B}} \rho_{\beta}(\tilde{p}(h) - q_{\beta}(h)).$$

Instance-adaptive scaling uses a conservative predicted success value p_L :

$$N_{\text{IAS}} = \min \left\{ N_{\text{max}}, \left\lceil \frac{\log(1 - C)}{\log(1 - p_L)} \right\rceil \right\}.$$

See Sections 4.1–4.2 [24].

Conditions and theory. The budget identity uses independent trials: $1 - (1 - p)^N \geq C$. The paper’s Theorem 1 considers an uncapped budget and a held-out calibration adjustment. Fitted quantile regression alone is not exact finite-sample conformal coverage. A cap can defeat the target confidence. The existence of a correct candidate does not guarantee that the reranker chooses it; $p_L = 0$ requires a maximum-budget convention.

Connection. A required calibrated-PRM baseline. Its uncertainty target is continuation success, not intrinsic validity. Our extension should test selected-candidate and simultaneous-prefix coverage, rather than assuming that marginal calibration survives search or policy updates.

2.7. Statistical foundations for constructing and calibrating a tube

Deep ensembles [30]. Model and equations. For Gaussian regression, member m minimizes $\frac{1}{2} \log \sigma_m^2(x) + \frac{(y - \mu_m(x))^2}{2\sigma_m^2(x)}$. The predictive mixture has moments

$$\bar{\mu} = \frac{1}{M} \sum_m \mu_m, \quad v = \frac{1}{M} \sum_m \sigma_m^2 + \frac{1}{M} \sum_m (\mu_m - \bar{\mu})^2.$$

The second term measures disagreement between fitted means; the first represents within-member predictive variation. These are mixture moments, not a claim that the mixture itself is Gaussian. The paper combines proper scoring rules with independently initialized networks; classification predictions average probabilities.

Conditions and evidence. Predictive calibration and robustness are empirical findings. Independent initialization does not establish independence of errors on new inputs, especially with shared training data or representations. An ensemble standard deviation is neither a finite-sample confidence radius nor a guarantee of coverage under optimization.

Connection. A practical starting estimator for the tube center and heterogeneous width. Retaining each member’s entire vector of prefix predictions also supplies coherent trajectory scenarios. Compare between-member uncertainty against total predictive variance: penalizing irreducible outcome randomness can reject valid but difficult reasoning. Independent calibration and attack tests remain necessary.

Self-normalized confidence sets for linear bandits [31]. Theorem and equations. With $Y_t = X_t^\top \theta_* + \eta_t$, predictable features, conditionally σ -sub-Gaussian noise, and $\|\theta_*\|_2 \leq S$, define

$$V_n = \lambda I + \sum_{t \leq n} X_t X_t^\top, \quad \hat{\theta}_n = V_n^{-1} \sum_{t \leq n} X_t Y_t.$$

Theorem 2 gives, simultaneously over n , with probability at least $1 - \delta$,

$$\left\| \hat{\theta}_n - \theta_* \right\|_{V_n} \leq \sigma \sqrt{2 \log \frac{\det(V_n)^{1/2}}{\det(\lambda I)^{1/2} \delta}} + \sqrt{\lambda} S =: b_n.$$

Thus prediction error is bounded by $b_n \|\phi(h)\|_{V_n^{-1}}$ on that event.

Conditions and scope. Adaptively chosen features are allowed, but the linear model, predictable design, noise condition, and parameter bound are substantive assumptions. A neural feature map learned on the same data, a misspecified linear head, or logistic preference likelihood requires additional analysis. OFUL uses the set for optimism, rather than pessimistic reward selection.

Connection. This is a rigorous template for a shared-parameter tube, with larger widths in poorly observed directions. AdvPO already adapts related geometry to reward optimization. A project contribution must address PRM-specific dependence, misspecification, or coverage, rather than rediscover this ellipsoid.

Conformalized quantile regression (CQR) [32]. Construction and guarantee. Fit lower and upper quantiles on a training split. On m held-out examples compute

$$S_i = \max\{\hat{q}_{\alpha/2}(X_i) - Y_i, Y_i - \hat{q}_{1-\alpha/2}(X_i)\}, \quad k = \lceil (m+1)(1-\alpha) \rceil.$$

Let q be the k th calibration order statistic, taking $q = +\infty$ if $k > m$. The interval is

$$C(x) = [\hat{q}_{\alpha/2}(x) - q, \hat{q}_{1-\alpha/2}(x) + q].$$

Exchangeability of calibration and test examples yields marginal coverage $\mathbb{P}\{Y_{\text{new}} \in C(X_{\text{new}})\} \geq 1 - \alpha$; the upper bound is $1 - \alpha + 1/(m+1)$ under appropriate absence of ties.

Conditions and limits. The quantile model need not be correct for marginal coverage, but affects efficiency. The guarantee does not assert coverage conditional on every prefix, simultaneously over a trajectory, or after reward-based selection. Binary targets can produce uninformative prediction sets.

Connection. CQR calibrates a future random label or return, not directly the unknown mean reward function. To obtain a trajectory tube, define a suitable trajectory-level score and calibration unit; do not simply reinterpret stepwise prediction intervals as simultaneous confidence bands.

JOMI: selection-conditional conformal prediction [33]. Construction and guarantee. For a selected test index j , JOMI forms candidate-label-dependent reference sets $\mathcal{R}_j(y)$ by swapping calibration and test examples and retaining swaps that preserve the relevant selection event. Schematically, its prediction set is

$$C_j = \{y : V(X_j, y) \leq Q_{1-\alpha}(\{V_i : i \in \mathcal{R}_j(y)\} \cup \{+\infty\})\}.$$

Under the paper’s permutation and exchangeability conditions, coverage holds conditional on j being selected and a specified class of selection outcomes. Randomization can handle ties and sharpen exactness.

Conditions and limits. The selection rule must satisfy the required invariance, and the reference set must actually implement that rule. The relevant calibration/test exchangeability is conditional on other test covariates. Candidates generated for one shared problem are dependent; treating every prefix or candidate as an independent calibration example does not establish this premise.

Connection. Especially relevant to best-of- N , where ordinary marginal coverage can fail at the selected maximum. Problem-level calibration or a justified conditional sampling model is needed. This covers the selected prediction target; it does not automatically give joint trajectory coverage or validity throughout an adaptive RL training loop.

Weighted conformal prediction under covariate shift [34]. Construction and guarantee. Suppose $Q_X \ll P_X$, the density ratio $w(x) = dQ_X/dP_X$ is known, and $P_{Y|X} = Q_{Y|X}$. Split conformal replaces the ordinary score quantile by the weighted distribution

$$\sum_{i=1}^m \frac{w(X_i)}{\sum_{k=1}^m w(X_k) + w(x)} \delta_{S_i} + \frac{w(x)}{\sum_{k=1}^m w(X_k) + w(x)} \delta_{+\infty}.$$

Its $(1 - \alpha)$ quantile yields marginal coverage for a test point drawn from Q , while calibration examples come from P .

Conditions and limits. The infinite test-point mass is essential to the finite-sample correction. Poor overlap can force an infinite quantile. Estimated or clipped weights do not inherit the exact theorem without accounting for their error. The theorem assumes stable conditional labels, not arbitrary joint distribution shift.

Connection. This offers a route from reference-policy calibration to a new policy’s prefix distribution. If a PRM target is continuation success, changing the rollout policy can change $Y | X$, violating covariate shift. Policy selection using calibration labels also needs separate treatment. Measure overlap and weight concentration before claiming a usable tube under optimization.

Conformal prediction beyond exchangeability [35]. Theoretical detail. The paper quantifies degradation using swapped-data laws rather than assuming exact exchangeability. A representative coverage-gap bound for prespecified weights $w_i \in [0, 1]$ has the form

$$(1 - \alpha) - \mathbb{P}\{Y_{n+1} \in C(X_{n+1})\} \leq \frac{\sum_{i=1}^n w_i \text{TV}(\mathcal{L}(Z), \mathcal{L}(Z^{(i)}))}{1 + \sum_{i=1}^n w_i},$$

where $Z^{(i)}$ swaps observation i with the test observation. The paper also handles nonsymmetric fitting procedures using its randomization construction.

Conditions and limits. The right side must be small to obtain a useful statement. Recency weighting can reduce the influence of old observations but also reduces calibration information. Data-dependent weights require the relevant conditioning or additional justification. Bounding swap distances for adaptively generated reasoning trajectories is a separate research problem.

Connection. Useful for analyzing gradual policy drift and periodically recalibrated PRMs. It supplies a degradation statement, not a guarantee under arbitrary reward-driven distribution shift. An experiment should report how coverage changes with policy divergence and calibration age, rather than silently assuming that a reference-policy conformal guarantee survives RL.

Conformal risk control (CRC) [36]. Construction and theorem. Let $L_i(\lambda)$ be exchangeable loss functions, bounded above by B , nonincreasing and right-continuous in λ . With $\hat{R}_n(\lambda) = n^{-1} \sum_i L_i(\lambda)$, choose

$$\hat{\lambda} = \inf \left\{ \lambda : \frac{n}{n+1} \hat{R}_n(\lambda) + \frac{B}{n+1} \leq \alpha \right\}.$$

With an admissible conservative endpoint and the paper’s conditions, $\mathbb{E}[L_{n+1}(\hat{\lambda})] \leq \alpha$.

Conditions and limits. This is an expected-risk guarantee, not a high-probability bound on every deployed instance or risk conditional on selection. Monotonicity concerns the actual loss function. If changing λ changes which trajectory is selected, the resulting error rate need not be monotone. Empty-set or abstention behavior must be included in the definition.

Connection. A suitable PRM application is thresholding fixed scores:

$$L_i(\lambda) = \mathbf{1}\{\text{invalid step}_i\} \mathbf{1}\{s_i \geq \lambda\}.$$

This loss is nonincreasing. This controls erroneous acceptance frequency, which differs from the fraction of accepted steps that are invalid. For a pessimism coefficient used inside best-of- N or RL, first prove the required monotonicity or use another calibration method.

CROMS: model selection for conformalized robust optimization [37]. Decision formulation. A conformal set induces the robust decision

$$z_\lambda(x) \in \arg \min_z \max_{y \in C_\lambda(x)} \varphi(y, z), \quad \lambda_* \in \arg \min_\lambda \mathbb{E}[\varphi(Y, z_\lambda(X))].$$

Thus model selection targets realized decision loss, rather than interval width alone. The paper distinguishes efficient E-CROMS, with approximate/asymptotic validity and optimality results, from a symmetric augmented-data F-CROMS construction with finite-sample coverage and asymptotic optimality. Its localized variant requires additional nonparametric assumptions.

Conditions and limits. Ordinary split-conformal coverage cannot simply be retained after arbitrarily choosing the model on the same calibration data. The guarantees depend on the specified selection construction and regularity conditions. Optimizing a narrow interval can select a model with worse robust decisions; conversely, conservative decisions can lose substantial nominal utility.

Connection. Particularly relevant when choosing among tube geometries, uncertainty estimators, or pessimism levels. Calibrate and select the entire uncertainty-to-decision pipeline, and evaluate the resulting reasoning success and abstention cost. Decision-aware calibration is already established here; novelty would require a PRM-specific selection/dependence analysis or demonstrably improved robust reasoning behavior.

2.8. Robust optimization and reward shaping

Distributionally robust optimization (Duchi–Namkoong) [40]. Objective and duality.

For loss ℓ_θ , optimize

$$\min_{\theta} \sup_{Q \ll P: D_f(Q||P) \leq \rho} \mathbb{E}_Q[\ell_\theta].$$

Under suitable duality conditions, its inner value can be expressed as

$$\inf_{\lambda \geq 0, \eta} \left\{ \lambda \rho + \eta + \lambda \mathbb{E}_P \left[f^* \left(\frac{\ell_\theta - \eta}{\lambda} \right) \right] \right\},$$

with boundary cases interpreted by limits. The paper develops guarantees for specified divergence families, including Cressie–Read divergences, under loss and statistical regularity assumptions.

Conditions and limits. Robustness to a subpopulation follows only when its conditional distribution lies inside the ambiguity set. A ball around an empirical distribution reweights observed support; it cannot create an unseen failure type. Divergence choice, radius, and tail assumptions determine how conservative and statistically estimable the objective is.

Connection. Distinguish uncertainty about the distribution of problems or trajectories from uncertainty about the reward function itself. They can be combined, but their adversaries and coverage events differ. A PRM project could use DRO for rare reasoning patterns while a reward-function tube handles evaluator error; identifying which uncertainty actually causes hacking is essential.

Robust dynamic programming (Iyengar) [41]. Structural principle. Rectangular transition ambiguity permits local adversarial choices compatible with dynamic programming. A standard state-action rectangular discounted recursion can be represented as

$$(Tv)(s) = \max_a \inf_{p \in \mathcal{P}_{s,a}} \left\{ r(s, a) + \gamma \sum_{s'} p(s') v(s') \right\}.$$

For bounded rewards, $\gamma < 1$, and suitable nonempty uncertainty sets, the local operator preserves the contraction structure needed for value iteration. The paper’s central contribution is the role of rectangularity in robust dynamic programming and associated game interpretations.

Conditions and limits. Rectangularity means the uncertainty set allows the relevant combinations of local choices. A single globally shared uncertain parameter typically couples states and violates that structure. Substituting independent local worst cases changes the adversary and can produce excessive pessimism. This displayed recursion illustrates the standard specialization; it is not a claim that every reward tube meets the paper’s assumptions.

Connection. A trajectory tube must specify whether errors can be chosen independently at every prefix or must arise from one coherent reward function. That choice determines both conservatism and computational tractability. Robust Bellman updates require an explicit rectangular formulation or a justified state augmentation, not merely an uncertainty interval attached to each step.

Potential-based reward shaping [42]. Theorem and telescoping identity. The shaping reward

$$F(s, a, s') = \gamma\Phi(s') - \Phi(s)$$

changes a finite-horizon discounted return by

$$\sum_{t=0}^{T-1} \gamma^t (r_t + F_t) = \sum_{t=0}^{T-1} \gamma^t r_t - \Phi(s_0) + \gamma^T \Phi(s_T).$$

With a fixed initial state and appropriate terminal boundary, or a bounded potential in an infinite discounted horizon, this preserves policy ordering. The necessity result concerns transformations that preserve optimal policies across the paper’s broad class of environments.

Conditions and limits. A poor but fixed potential can still preserve ordering when used consistently; it need not accelerate learning. Terminal potentials and discount conventions matter. Arbitrary positive process rewards, stepwise probability products, or min-form returns are not automatically potential-based shaping.

Connection. If a PRM estimates a reasoning potential, uncertainty should be placed on a shared potential function and propagated through differences. Independently minimizing each difference can break telescoping by choosing incompatible endpoint values. This provides a precise way to distinguish uncertainty in a policy-invariant shaping signal from uncertainty in a new objective that can alter the optimal reasoning strategy.

2.9. Benchmarks and what their labels can establish

ProcessBench [43]. Task and metric. The benchmark contains approximately 3,400 reasoning traces, with the earliest erroneous step or a no-error label. Its reported harmonic metric combines exact first-error localization accuracy on erroneous traces and no-error accuracy on clean traces:

$$F = \frac{2A_{\text{err}}A_{\text{clean}}}{A_{\text{err}} + A_{\text{clean}}}.$$

This is a harmonic mean of two accuracies, rather than ordinary stepwise binary classification F1. PRM evaluation thresholds are chosen on a designated subset in the original protocol.

Evidence and limits. The benchmark tests mathematical error detection across difficulty levels. Correct final answers can coexist with invalid intermediate reasoning. Earliest-error annotation does not supply verified labels for every later step, so it cannot directly establish simultaneous all-step coverage. A static localization score also does not measure robustness to an optimizer searching for high-scoring wrong solutions.

Connection. Use it to test error localization and clean-trace rejection, with thresholds and uncertainty calibration fixed independently of evaluation. Report false acceptance and abstention alongside the benchmark metric. For a full trajectory-tube coverage claim, obtain all-step labels or define a coverage event restricted to the observed first-error target.

PRMBench [44]. Task and metric. PRMBench reports 6,216 instances and 83,456 step labels, organized around simplicity, soundness, and sensitivity, with nine finer dimensions. Its PRM-score

combines negative-class and overall F1:

$$\text{PRMscore} = w_1 F_{1,\text{neg}} + w_2 F_{1,\text{overall}}.$$

The benchmark probes redundant or circular reasoning, correctness failures, and sensitivity to changes in reasoning. The weighting and threshold protocol should be recorded when reproducing results.

Evidence and limits. Step-level discrimination reveals weaknesses hidden by aggregate final-answer accuracy. However, redundancy is a utility or efficiency judgment, not necessarily mathematical invalidity. Aggregating these dimensions into a single binary target can obscure what a PRM’s reward represents. Static test cases do not certify calibration under adversarial optimization.

Connection. Separate correctness, informativeness, and sensitivity in our labels and reported results. A tube designed for correctness should not be judged solely by its agreement with a simplicity preference. Paired transformed traces can test whether uncertainty responds to semantic errors while remaining stable under harmless rewrites; that invariance diagnostic is a proposed project evaluation, rather than a coverage theorem supplied by PRMBench.

Socratic-PRMBench [45]. Task and metric. The benchmark has 2,995 samples spanning transformation, decomposition, regathering, deduction, verification, and integration, with twenty error types. Its aggregate score is explicitly

$$\text{Score} = \frac{1}{2} F_{1,\text{neg}} + \frac{1}{2} F_{1,\text{overall}}.$$

The construction uses structured reasoning patterns, synthetic modifications, and filtering, with a sampled human assessment of annotation agreement.

Evidence and limits. Pattern-level results can reveal that a reward model recognizes deduction errors but misses verification or integration failures. Sampled annotation agreement does not certify every synthetic label. Aggregate F1 can conceal a high false-acceptance rate in a rare pattern and says nothing directly about coverage on optimizer-selected traces.

Connection. A useful stress test for heterogeneous tube widths and structured uncertainty. Our proposed diagnostic is empirical coverage and false acceptance within each prespecified pattern, together with interval width and abstention. Such subgroup results require enough independent problems per group; they do not imply unrestricted conditional coverage for arbitrary prefixes. Pair this static benchmark with best-of- N and RL stress tests so that uncertainty quality is evaluated where reward hacking actually appears.

2.10. Synthesis and research positioning

Family	Established capability	What our study must resolve
Process supervision	Intermediate verification	What quantity the tube covers
Label denoising	Better synthetic labels	Coverage after optimization
PURE / shaping	Better credit assignment or outcome invariance	Tube benefits with aggregation held fixed

Family		Established capability	What our study must resolve
Ensembles / Ad- vPO		Pessimistic reward optimization	Shared bias and joint process uncertainty
Quantile / OT PRMs		Prefix distributions and bounds	Selection coverage and completer dependence
Conformal / JOMI		Guarantees for specified sampling rules	Feasible PRM adaptation and adaptive RL assumptions
Correlated-proxy robustness		Structured worst-case rewards	Construction from process supervision

2.10.1 A defensible candidate contribution

Optimization-aware uncertainty for process rewards: compare pointwise intervals with structured or problem-level tubes; measure coverage after selection; determine whether coverage improvements reduce hacking at matched optimization pressure. Novelty remains provisional. An empirical paper could expose systematic post-selection undercoverage and a practical repair. A theory paper could characterize safe improvement and failure of marginal intervals. A combined paper would connect a restricted theorem to measured violations in real PRMs. No publication outcome is assumed.

Part II

Notes: Foundations for the Project

3. Notation and the objectives we must keep separate

Symbol	Meaning
q	Problem or prompt drawn from a task distribution
s_t, h_t	Reasoning step and prefix $h_t = (q, s_1, \dots, s_t)$
τ, T	Complete trajectory and number of reasoning steps
π_θ, π_0	Policy being optimized and fixed reference policy
z_t	State before action a_t ; in text, the prompt and current prefix
c_t, v_t	Local step-correctness and cumulative prefix-validity labels
$p_{\pi_c}(h_t)$	Probability a fixed completer π_c finishes correctly
$\hat{r}_\phi, r^*$	Learned process reward and specified target process reward
G	Explicit step-to-trajectory aggregation rule
$\hat{R}, R^*$	Proxy trajectory reward and intended task utility
$\mathcal{U}, L, U$	Reward uncertainty set and its lower and upper envelopes
b, α, δ	Width; prediction miscoverage level; confidence-set failure probability
ρ, β	Ambiguity-set radius and policy KL regularization coefficient

Critical distinction. In general,

$$c_t \neq v_t \neq p_{\pi_c}(h_t), \quad G(r^*(h_1), \dots, r^*(h_T)) \neq R^*(\tau).$$

A valid algebraic manipulation can follow an earlier false assumption; a wrong step can later be repaired; a correct prefix can be difficult for a weak model to complete. Even perfectly estimating step correctness does not automatically make an arbitrary aggregation equal final-answer accuracy.

4. Reinforcement learning in the reasoning setting

4.1. MDPs, states, actions, and policies

A finite-horizon Markov decision process specifies states, actions, transition probabilities, rewards, an initial-state distribution, and a horizon. The Markov property says that, conditional on the current state and action, the next-state distribution does not additionally depend on earlier history:

$$\mathbb{P}(z_{t+1} \mid z_{0:t}, a_{0:t}) = P(z_{t+1} \mid z_t, a_t).$$

It does not mean only the immediately preceding observation matters: the state can encode the entire history. Text generation is Markov when the state includes the full prompt and generated prefix. An action can be one token, one reasoning step, or a tool call. Appending text is deterministic given the action; sampling actions from the policy is stochastic. Tool environments may also have stochastic or hidden state. A partially observable formulation is needed when observations do not determine the environment state.

A stochastic policy maps a state to an action distribution. A deterministic policy picks one action. A stationary policy depends on the current state, and a nonstationary one may also depend on time. Time can be appended to the state in a finite-horizon problem. Existence of stationary deterministic optima requires the usual MDP conditions and objective; do not assert it for arbitrary partial observability or nonlinear robust objectives.

4.2. Returns, value, and credit assignment

Use the convention that r_t is earned by action a_t :

$$G_t^{\text{sum}} = \sum_{k=t}^{T-1} \gamma^{k-t} r_k, \quad V^\pi(z) = \mathbb{E}_\pi[G_t^{\text{sum}} \mid z_t = z], \quad Q^\pi(z, a) = \mathbb{E}_\pi[G_t^{\text{sum}} \mid z_t = z, a_t = a].$$

The advantage $A^\pi(z, a) = Q^\pi(z, a) - V^\pi(z)$ measures whether an action is better than the policy's average action. For additive returns,

$$Q^\pi(z, a) = \mathbb{E}[r_t + \gamma V^\pi(z_{t+1}) \mid z_t = z, a_t = a].$$

Sparse terminal rewards make it hard to determine which earlier decisions caused success. Dense process feedback may help, but repeated erroneous rewards also accumulate. The classical Monte Carlo, temporal-difference, value-learning, and actor-critic vocabulary is useful background; these notes use a finite episodic reasoning model rather than reproduce all tabular algorithms. The textbook is a background reference [46].

4.3. Policy gradients, PPO, and GRPO

For a fixed trajectory reward, the likelihood-ratio identity gives

$$\nabla_\theta \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)] = \mathbb{E} \left[R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t \mid z_t) \right].$$

Baselines reduce variance; return-to-go or advantage estimates improve credit assignment. A critic estimates value; it is a separate source of approximation error from the reward model.

PPO commonly uses the clipped surrogate [47]

$$\mathbb{E}_t \left[\min\{\omega_t(\theta) \hat{A}_t, \text{clip}(\omega_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t\} \right], \quad \omega_t(\theta) = \frac{\pi_\theta(a_t \mid z_t)}{\pi_{\text{old}}(a_t \mid z_t)}.$$

Clipping limits this surrogate's incentive for large individual updates; it is not a hard guarantee that policy KL remains small. The policy can still move far from its reference through many small updates.

GRPO estimates relative advantages from groups of sampled answers, avoiding a separately learned value critic in its common formulation [48]. A simple outcome-level version uses

$$\hat{A}_j = \frac{R_j - \bar{R}}{\text{std}(R_1, \dots, R_m) + \varepsilon}.$$

Dense process rewards require a stated token/step assignment; broadcasting one trajectory advantage to every token is not stepwise process credit assignment. Group normalization can obscure raw reward scales and gives little signal when all group rewards coincide. Document the actual variant in experiments.

4.4. RLHF, RLVR, and inference-time search

RLHF commonly trains a preference reward model and optimizes against it. RLVR uses automatically checked outcomes, such as answers or test results. The checker is only as reliable as its specification and implementation. PRMs can provide dense supervision in either setting. Best-of- N changes which sampled trajectory is selected while keeping generator weights fixed:

$$\tau_{\text{sel}} = \arg \max_{j \leq N} \hat{R}(\tau_j), \quad \tau_j \sim \pi_0.$$

Beam search additionally selects prefixes and changes the effective exploration distribution. RL changes generator parameters, allowing repeated adaptation to the reward’s errors. Results in one regime need not transfer to another.

5. What a process reward model does

A discriminative PRM predicts correctness scores at step boundaries from the question and preceding reasoning. A generative PRM emits critiques or labels. Conditioning on preceding steps is common; “stepwise” does not mean the model ignores all previous context. The reward head may output a logit, a probability, or a free scalar. The domain and scale determine admissible uncertainty sets.

5.1. Labels and training

For binary labels, a standard loss is

$$\ell_{\text{BCE}}(\phi; h_t, y_t) = -y_t \log \hat{r}_\phi(h_t) - (1 - y_t) \log(1 - \hat{r}_\phi(h_t)).$$

Human annotation may stop at the first error. Later steps then have missing or censored labels; treating them all as independently labeled local errors changes the target. Automated judges introduce correlated annotation mistakes.

For K independent completions under a fixed completer π_c ,

$$\tilde{p}(h_t) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}\{\text{completion } k \text{ succeeds}\}, \quad \text{Var}(\tilde{p} \mid h_t) = \frac{p_{\pi_c}(1 - p_{\pi_c})}{K}.$$

A label for “at least one success” has mean $1 - (1 - p_{\pi_c})^K$, not p_{π_c} . It changes with K and can approach one for any nonzero success probability. Model repair makes completion success different from correctness of the present step. Thus Monte Carlo uncertainty and semantic label mismatch are distinct problems.

5.2. Aggregation creates additional incentives

For probability-like scores $r_t \in [0, 1]$, common choices are

$$G_{\text{sum}} = \sum_t r_t, \quad G_{\text{mean}} = T^{-1} \sum_t r_t, \quad G_{\text{min}} = \min_t r_t, \quad G_{\text{prod}} = \prod_t r_t.$$

Sum rewards repetition of easy steps. Mean can dilute a bad step with easy ones. Minimum emphasizes the weakest step but can reward short or vacuous responses and rejects repair unless repair is explicitly modeled. Product penalizes length: $0.95^{10} \approx 0.60$ and $0.95^{30} \approx 0.21$. It is a joint-correctness probability only under an appropriate probabilistic interpretation; multiplying ordinary marginal step scores does not justify an independence claim.

For bounded per-step perturbations $|r_t - \tilde{r}_t| \leq \epsilon$, sum errors can grow as $T\epsilon$, while minimum errors are bounded by ϵ . This deterministic observation concerns the aggregation target, not proof that minimum rewards optimize true mathematical accuracy. Splitting or merging steps can change all four scores. Include segmentation attacks and empty-output rules.

6. Why reward hacking occurs and why it matters

Write $\hat{R}(\tau) = R^*(\tau) + e(\tau)$ schematically. Maximizing $\hat{R}$ selects both genuine quality and positive error. Error on ordinary data is not the same as error among the candidates chosen by the optimizer.

6.1. A numerical example

Suppose a correct solution scores 0.80 while an incorrect fluent solution scores 0.95. A policy increasing the latter’s probability improves proxy reward and reduces correctness. A useful tube might give the correct solution $[0.76, 0.84]$ and the exploitable solution $[0.20, 1.00]$, making pessimistic selection prefer the correct solution. If both intervals are too narrow, or the wrong solution’s uncertainty is underestimated, this defense fails. If every interval is $[0, 1]$, the lower score gives no useful ranking.

6.2. Different mechanisms need different diagnostics

- **Selection of noise:** higher search budgets find larger positive errors even without new policy weights.
- **Distribution shift:** optimization visits reasoning prefixes absent from training, where feature–correctness associations fail.
- **Misspecification:** a scorer rewards length, style, or easy local claims instead of the intended reasoning property.
- **Objective mismatch:** perfectly estimated local rewards are summed or transformed into a trajectory objective that rewards the wrong behavior.
- **Evaluator manipulation:** delimiters, injected instructions, or checker loopholes change what the evaluator measures.

Optimization does not require conscious intent to exploit a reward. Conversely, an ordinary wrong answer is not by itself evidence of hacking. Look for reward improvement coupled to independent quality stagnation or decline and a mechanism that explains it. Hacking harms model quality, invalidates capability estimates, wastes compute, and can train persistent undesirable behavior.

7. Uncertainty, calibration, and coverage

7.1. Three kinds of uncertainty

Epistemic uncertainty concerns limited knowledge of the target reward function. **Aleatoric variability** concerns stochastic outcomes or ambiguous labels given the input. **Misspecification** means the model class or label definition systematically misses relevant truth. Ensemble variance may detect the first while missing the third. Rollout variance primarily measures outcome variability and finite-sample noise, not intrinsic logical ambiguity.

7.2. Prediction intervals are not confidence bands for a function

An interval covering a new binary label is a prediction set for Y . A confidence interval for $p(x) = \mathbb{P}(Y = 1 \mid X = x)$ covers a conditional mean. A function confidence set asserts simultaneous containment of p over a domain. They require different information and assumptions. A binary-label interval that contains both zero and one is valid but gives little reward discrimination. Quantiles of observed labels or rollout estimates do not automatically become confidence bounds on intrinsic correctness probabilities.

Calibration of a probability score μ means that among examples assigned a score near u , the event frequency is near u . This is distribution-dependent. Low aggregate expected calibration error can coexist with poor accuracy on rare high-scoring mistakes. Ranking accuracy and calibration are also different.

Marginal coverage is

$$\mathbb{P}\{Y_{\text{new}} \in C(X_{\text{new}})\} \geq 1 - \alpha.$$

Trajectory coverage contains all relevant steps. Candidate-set coverage contains all trajectories being searched. Selection coverage concerns the chosen one. Function-level coverage contains the target reward on every point of a stated domain. Repeated RL adaptation requires an event valid for all encountered queries or a sequentially justified update procedure; a static marginal claim does not provide it.

8. Robust optimization: the geometry matters

For $\hat{r} : \mathcal{H} \rightarrow [0, 1]$, a pointwise box tube is

$$\mathcal{U}_{\text{box}} = \{r : |r(h) - \hat{r}(h)| \leq b(h), h \in \mathcal{H}\}.$$

It allows each point's error to vary independently. A function set induced by one uncertain parameter w couples different prefixes. An adversary choosing one reward function for the entire policy gives

$$\max_{\pi} \inf_{r \in \mathcal{U}} \mathbb{E}_{\tau \sim \pi} [G(r(h_1), \dots, r(h_T))].$$

An adversary choosing a different model after each trajectory gives

$$\max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\inf_{r \in \mathcal{U}} G(r(h_1), \dots, r(h_T)) \right].$$

Generally the second is no larger than the first. They can coincide for certain rectangular sets but need not for coupled sets. State explicitly which is solved.

Distributional robustness instead permits an adversary to change a distribution. For robust training this might be

$$\min_{\phi} \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\ell_{\text{PRM}}(\phi; h, Y)].$$

It changes the fitted model, whereas policy pessimism changes how a fixed model is used. Statistical DRO over examples, DRO over uncertain reward values, and robust control over functions should not share one undifferentiated symbol.

9. Distribution movement and reward shaping

For $\gamma < 1$, define normalized discounted occupancy

$$d_{\pi}(z, a) = (1 - \gamma) \mathbb{E}_{\pi} \sum_{t \geq 0} \gamma^t \mathbf{1}\{z_t = z, a_t = a\}.$$

Additive expected return is $(1 - \gamma)^{-1} \langle d_{\pi}, r \rangle$. For finite-horizon reasoning with $\gamma = 1$, use the finite sum of state-action visitation measures instead. Policy action KL, trajectory KL, and occupancy divergence are different quantities. Under shared dynamics, trajectory KL can be expressed as the expected sum of conditional action KLs, but an empirically averaged token KL still needs normalization and sampling conventions.

An error bound averaged under d_{π_0} may be amplified under d_{π} . Importance weighting requires $d_{\pi} \ll d_{\pi_0}$ and a usable density ratio; theoretical support overlap can be practically useless if weights are enormous.

With a reliable base task reward, potential shaping uses

$$F_t = \gamma \Phi(z_{t+1}) - \Phi(z_t).$$

The discounted sum telescopes:

$$\sum_{t=0}^{T-1} \gamma^t F_t = -\Phi(z_0) + \gamma^T \Phi(z_T).$$

Setting the terminal potential to zero removes dependence on chosen actions [42]. Fix the potential during each evaluated episode and handle truncation explicitly. Arbitrary dense PRM bonuses, adaptive within-episode potentials, or nonzero length-dependent terminal potentials do not inherit this argument.

10. Research habits that make the results interpretable

Fix the target and label protocol before selecting an uncertainty estimator. Split data by problem, retain independent calibration and evaluation sets, and keep attack construction separate from final transfer tests. Report correctness, process validity, uncertainty coverage, computational cost, and reward curves. An uncertain reward estimator cannot establish its own success merely by giving higher scores. The detailed experiment plan and decision rules follow in Part III.

Part III

Constructing and Characterizing the Reward Tube

11. From the line metaphor to a research object

Along a trajectory, the PRM produces the line $\hat{r}(\tau) = (\hat{r}(h_1), \dots, \hat{r}(h_T))$. A tube should contain plausible target rewards, not merely decorate that line with arbitrary error bars. Define lower and upper envelopes $L(h), U(h)$, a domain $\mathcal{H}_B$ indexed by an optimization budget B , and a set of functions

$$\mathcal{U}_B = \{r : \mathcal{H}_B \rightarrow [0, 1] : L(h) \leq r(h) \leq U(h), \quad r \text{ satisfies specified coupling constraints}\}.$$

The coupling says which combinations of step errors are simultaneously plausible. Independent intervals form a box; shared parameters form a correlated tube. This distinction is the main geometric research question.

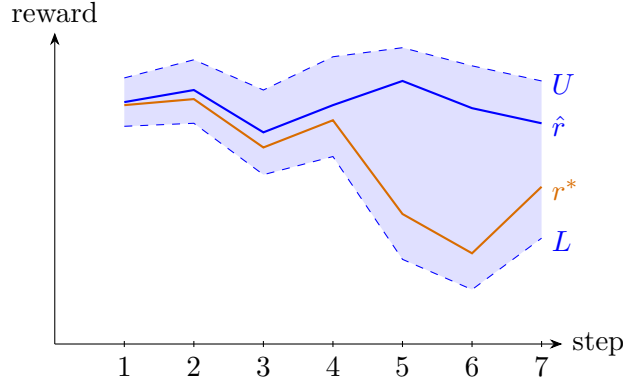


Figure 1: Illustrative tube with heterogeneous width. All values are schematic, not measured results. Drawing a target line inside the band does not establish statistical coverage or describe which correlated functions are admissible.

11.1. The design specification

Before implementing any method, specify:

1. **Target:** human/prover step validity, cumulative prefix validity, fixed-completer success probability, or task utility. We recommend intrinsic step/prefix validity as the primary PRM target and independent answer correctness as the task metric. Use a separate fixed-completer track for rollout calibration.
2. **Domain:** a finite candidate pool, trajectories from a frozen policy, a bounded policy family, or all feature vectors in a known domain.
3. **Geometry:** independent box, shared parameter ellipsoid, low-rank bias set, budgeted deviations, or a joint reward distribution.

4. **Assurance:** empirical coverage, marginal prediction coverage, candidate-set coverage, uniform function coverage, or a policy improvement bound.
5. **Use:** robust reranking, trajectory-level RL, robust policy-level improvement, or task-preserving shaping. These are different objectives.

The methods below are proposals. Their novelty and effectiveness are hypotheses.

12. Approach A: problem-level calibration of candidate tubes

Purpose. Establish a practical baseline and diagnose whether selection breaks ordinary coverage. This is an empirical route with a modest, explicit exchangeability guarantee for a frozen candidate-generation protocol.

12.1. Construction

Train a center $\mu(h) \in [0, 1]$ and positive scale $s(h) \geq s_{\min} > 0$ using only training problems. Scale candidates include ensemble disagreement, last-layer leverage, predicted residual magnitude, or quantile widths. Fix the scorer, generator, parser, and maximum budget B before calibration. For each calibration problem i , generate all B candidates and collect target labels y_{ijt} for their steps. Problems, not individual steps, are the exchangeable calibration units. Define

$$S_i^- = \max_{j \leq B, t \leq T_{ij}} \frac{\mu(h_{ijt}) - y_{ijt}}{s(h_{ijt})}, \quad S_i^+ = \max_{j \leq B, t \leq T_{ij}} \frac{y_{ijt} - \mu(h_{ijt})}{s(h_{ijt})}.$$

On m calibration problems, let $q^\pm$ be the $[(m+1)(1-\alpha_\pm)]$ th order statistic of $S_i^\pm$, augmented with $+\infty$ if the index exceeds m . Replace negative quantiles by zero to make widths nonnegative. Set $\alpha_- + \alpha_+ = \alpha$ and

$$L(h) = \max\{0, \mu(h) - q^- s(h)\}, \quad U(h) = \min\{1, \mu(h) + q^+ s(h)\}.$$

A one-sided version spends all the error budget on S^- and reports only a lower prediction bound; it must not be called a two-sided confidence tube.

Under exchangeable problem blocks, independent model fitting, and the same frozen generation/labeling protocol, split-conformal ranks and a union bound give

$$\mathbb{P}\{L(h_{jt}) \leq y_{jt} \leq U(h_{jt}) \text{ for all candidates and steps in a new block}\} \geq 1 - \alpha.$$

This adapts standard conformal reasoning [32]; it is not a new theorem. Within-block steps and candidates can be dependent. Any reranker choosing from that fixed block inherits this containment event. The probability is marginal over calibration and new problem blocks, not conditional coverage for every problem or a confidence band over the whole reward function.

12.2. The target problem, and how to handle it

If y is a binary validity label, this construction covers the *realized label*. Many honest intervals may reach zero, making lower-bound selection uninformative. That is a result to report, not something to

fix by relabeling the interval as epistemic uncertainty. Use sets of valid/invalid labels or abstention for this track; evaluate informativeness through singleton-set frequency and false acceptance.

For a success-probability track, keep completer π_c fixed and label each prefix with K independent completion outcomes. Calibration then covers future rollout estimates $\tilde{p}$, not p_{π_c} . If the new block has at most $M = BT_{\max}$ prefixes, Hoeffding and a union bound give an additional allowance

$$\eta_K = \sqrt{\frac{\log(2M/\delta_{\text{MC}})}{2K}}.$$

Expanding each interval by η_K covers the true fixed-completer success probabilities in that block with probability at least $1 - \alpha - \delta_{\text{MC}}$. Independent Bernoulli rollouts, a correct answer checker, and a fixed finite cap are assumptions. This does not convert success probabilities into intrinsic step correctness.

12.3. Using the tube

For the rectangular set and a coordinatewise nondecreasing aggregation G , rerank using

$$G(L(h_1), \dots, L(h_T)).$$

Repeat for nominal, calibrated-center, and uncertainty-penalized baselines with the same G . A candidate-wise lower reward may be overly conservative; it should be compared with a shared-function set in Approach B.

12.4. Tests, limitations, and novelty

Sweep nested fixed pools with $N \leq B$. Compare stepwise calibration, trajectory-block calibration, full candidate-block calibration, a union-bound correction, and a selection-conditional/JOMI adaptation [33]. Calibrating the selected candidate directly requires the selection protocol to be fixed without circular reuse of its calibration labels.

Problem blocks from an RL-updated generator or a beam search influenced by the new calibrated reward do not automatically match the frozen-pool guarantee. Block maxima can be too conservative and annotation cost scales with candidate count. A publishable result would need a useful improvement in selected-path coverage and correctness at matched cost, or a clear failure characterization. The max-score conformal construction itself is standard.

13. Approach B: a coupled function tube from frozen features

Purpose. Model one shared source of reward error across steps and policies, providing the cleanest initial theoretical route. AdvPO is close prior work [22]; the proposed extension is process coupling and validation.

13.1. A tractable model and an exact induced tube

Start with a frozen feature map $\psi(h) \in \mathbb{R}^d$ and linear real-valued reward $r_w(h) = \psi(h)^\top w$. Fit a regularized head $\hat{w}$ and let

$$V = \lambda I + \sum_{i=1}^n \psi_i \psi_i^\top, \quad \mathcal{C} = \{w : \|w - \hat{w}\|_V \leq \beta_n\}, \quad \mathcal{U} = \{r_w : w \in \mathcal{C}\}.$$

Here $\|u\|_V = \sqrt{u^\top V u}$. The induced pointwise envelope is

$$\psi(h)^\top \hat{w} \pm \beta_n \sqrt{\psi(h)^\top V^{-1} \psi(h)}.$$

Take $\lambda > 0$ so that V is invertible. This is a projection of one ellipsoid; arbitrary simultaneous choices of the pointwise endpoints are not allowed.

For $y_i = \psi_i^\top w^* + \varepsilon_i$, predictable frozen features, conditionally R_ε -sub-Gaussian noise, $\|w^*\| \leq S$, and regularized least squares, a standard confidence choice is

$$\beta_n = R_\varepsilon \sqrt{2 \log \left(\frac{\det(V)^{1/2}}{\det(\lambda I)^{1/2} \delta} \right)} + \sqrt{\lambda} S.$$

Self-normalized theory supplies adaptive-data confidence under these assumptions [31]. This does not justify an unrestricted neural PRM guarantee. Binary conditional means can fit this model only if the linear mean is correctly specified. Labels systematically supplied by a biased judge violate it.

For probability outputs $\text{sigmoid}(\psi^\top w)$, monotonicity yields sigmoid transforms of the logit envelopes *if* the parameter confidence set is valid. Least-squares confidence theory does not automatically validate a logistic head. Hessian/Laplace ellipsoids can be empirical baselines; a formal generalized-linear analysis must establish its own assumptions and radius.

13.2. Joint geometry and robust scoring

For a trajectory feature matrix Ψ_τ whose rows are $\psi(h_t)^\top$, the induced covariance shape is

$$\Sigma_\tau = \Psi_\tau V^{-1} \Psi_\tau^\top.$$

Repeated steps share uncertainty rather than generating new independent errors. For a linear aggregation $a^\top \mathbf{r}$, where a is fixed for the trajectory, the exact worst-case score for the unrestricted linear head is

$$\inf_{w \in \mathcal{C}} a^\top \Psi_\tau w = a^\top \Psi_\tau \hat{w} - \beta_n \sqrt{a^\top \Sigma_\tau a}.$$

This is a support-function identity, not a novel PRM theorem. If rewards must stay in $[0, 1]$, either ensure the ellipsoid satisfies the domain bounds or intersect it with those constraints and solve the resulting constrained problem. Clipping outputs or using sigmoid rewards generally destroys the linear identity. For a minimum objective, $\inf_w \min_t r_w(h_t) = \min_t \inf_w r_w(h_t)$; for products and nonlinear means, simultaneous endpoint substitution need not be exact.

Policy-level pessimism differs from per-trajectory pessimism. With $\bar{\psi}_\pi = \mathbb{E}_\pi \sum_t a_t \psi(h_t)$, the

former equals

$$\bar{\psi}_\pi^\top \hat{w} - \beta_n \|\bar{\psi}_\pi\|_{V^{-1}},$$

while the latter subtracts $\beta_n \mathbb{E}_\pi \|\sum_t a_t \psi(h_t)\|_{V^{-1}}$. Jensen makes the latter more conservative. Report which objective is optimized.

13.3. Misspecification and practical construction

An honest extension is

$$r^*(h) = \psi(h)^\top w^* + m(h), \quad |m(h)| \leq b_{\text{mis}}(h).$$

The resulting set couples parameter error but allows specified residual error. Estimate residual scales on separate problems containing adversarial and on-policy traces. Calling that estimate a uniform bound still requires a coverage argument. A residual floor can protect against zero ensemble variance, but should be evaluated through failures rather than assumed effective.

Implement frozen representations, ridge heads, then independently trained low-rank heads or adapters. Use Cholesky solves rather than explicitly forming V^{-1} . Training needs feature statistics; a full $d \times d$ matrix and quadratic-form scoring can be expensive. Compare dimension reduction, diagonal approximations, and low-rank approximations against full geometry. Do not hide the cost of computing PRM features.

13.4. Research payoff and risks

Compare a shared ellipsoid with its rectangular envelope at matched marginal width and matched joint coverage. Test common-mode bias, repeated steps, and rare logical transformations. The contribution could be a characterization of when coupling avoids excessive conservatism or controls policy exploitation. The main risks are feature misspecification, falsely small radii, and head-only diversity missing shared backbone errors. A generic last-layer ellipsoid is already covered by AdvPO and confidence-set literature.

14. Approach C: structured nuisance and budgeted error tubes

Purpose. Cover systematic shared biases that a seed ensemble misses. This route combines empirical attacks with an interpretable robust set.

Let $g(h)$ describe validated nuisance directions: step duplication, verbosity, unjustified certainty, answer-format markers, or question–solution mismatch. The last is a corruption detector, not a semantics-preserving nuisance. Define

$$r_{u,e}(h) = \hat{r}(h) - g(h)^\top u - e(h), \quad \|u\|_M \leq \rho_b.$$

For a probability target, intersect these constraints with $0 \leq r_{u,e}(h) \leq 1$; the unconstrained penalties below are then conservative bounds and need not be the exact constrained optimum. Alternatively, define the set on real-valued logits and optimize with the explicit link function. One u applies to the full trajectory or policy; specify that scope. Derive g from paired transformations whose labels are checked, not arbitrary text embeddings assumed to represent causality. CoLD motivates the

known-bias case [11]. Put a zero-centered uncertainty set around unexplained effects; an estimated known bias should first correct the center.

For nonnegative aggregation weights, an additional budgeted residual model is

$$|e_t| \leq b_t, \quad \sum_t \frac{|e_t|}{b_t} \leq \Gamma, \quad 0 \leq \Gamma \leq T,$$

omitting zero-width coordinates. Its worst-case linear penalty is the sum of the largest $\lfloor \Gamma \rfloor$ values of $a_t b_t$, plus the fractional part times the next largest. This standard robust-set geometry interpolates between no residual penalty and a rectangular all-steps penalty.

Construction protocol. Create paired correct rewrites and validated logical corruptions on training problems; fit bias features and residual scales; estimate a problem-level deviation budget on a disjoint calibration split; compare an independent box, a shared bias term, and a combined set. If a repeated systematic mistake affects every step, a small Γ is invalid. Use the shared bias component for such common modes and audit the remaining budget assumptions.

Potential contribution. Show that correlated nuisance uncertainty covers known failure directions at smaller useful widths than an unstructured box, then test held-out attacks. Compare against CoLD and hard-negative retraining at equal label cost. Primary risks are unreliable semantic validation and attacks outside g 's span. Token-count penalties alone are a baseline, not the new method.

15. Approach D: a joint reward distribution and robust risk

Purpose. Preserve dependencies in a distributional PRM rather than attach independent distributions to each step. This is an optional extension after simpler tube methods work; scalar distributional pessimism is prior work [27, 25, 28].

Fit a joint predictive distribution $p(\mathbf{r} \mid \tau)$ using shared latent parameters, low-rank factors, or ensemble trajectories. For a specified G , consider a reward-distribution ambiguity set

$$\mathcal{Q}_\rho(\tau) = \{Q : D_{\text{KL}}(Q \| p(\mathbf{r} \mid \tau)) \leq \rho\}.$$

Its robust expected aggregate is

$$\inf_{Q \in \mathcal{Q}_\rho} \mathbb{E}_Q G(\mathbf{r}) = \sup_{\eta > 0} \{-\eta \log \mathbb{E}_p e^{-G(\mathbf{r})/\eta} - \eta \rho\},$$

under the standard duality/integrability conditions, with boundary limits included where needed. The penalized form with fixed η omits $-\eta \rho$; it is a different problem. This variational identity is established robust-risk machinery, not a novel result.

For real-valued Gaussian rewards and linear $G = \mathbf{a}^\top \mathbf{r}$, the penalized score is $\mathbf{a}^\top \boldsymbol{\mu} - (\mathbf{a}^\top \Sigma \mathbf{a}) / (2\eta)$. Correlation changes the penalty; dropping off-diagonal terms can severely underestimate shared error. A Gaussian distribution is not a valid bounded probability distribution unless transformed or truncated, in which case the simple expression need not hold. Monte Carlo log-sum-exp estimates must be checked for tail-sampling error.

Calibration and experiments. Train distributional heads on repeated rollouts in the fixed-completer track or repeated independent annotations in the validity track, accounting for what those

repetitions measure. Compare independent versus joint models, entropic risk versus quantile bounds, and conservative distributions versus the same center without penalties. Measure tail fit after selection and shared errors. Distributional variability alone is not epistemic confidence; a perfectly fitted wrong label mechanism remains wrong. Publishability would require a process-specific gain, not re-deriving scalar entropic rewards.

16. Approach E: optimize, audit, and update the tube

Purpose. Make uncertainty estimation respond to the distribution created by reward optimization. This is the highest-cost combined route.

Define an optimization regime by initial policy, allowed updates, KL/length caps, algorithm, and budget. A reachable family Π_B induces prefix or trajectory distributions, rather than an unspecified ball around the training data. Empirically sample a mixture of checkpoints, best-of- N winners, beam prefixes, and attack-generated traces. A mixture covers its sampled modes; it does not certify every reachable policy.

1. Fit the initial PRM and set on training problems. Freeze its parameters and calibration during a bounded policy-update phase.
2. Optimize the robust score and separately attack it. Select audit examples with high robust score, low ensemble disagreement despite suspicious reasoning, and coverage failures; include random on-policy examples to avoid blind sampling.
3. Obtain independent validity labels or fixed-completer rollout targets. Keep a fresh evaluation stream untouched by reward updates.
4. Retrain the center or uncertainty model, then recalibrate on fresh data. Compare with equal-budget center-only retraining.
5. Validate new policies after the update; use a predefined accept/defer rule when evidence of true improvement or valid coverage is insufficient.

If policy π_k is fixed before fresh calibration and test blocks are drawn under it, a per-round guarantee conditional on the previous history can be combined with $\sum_k \alpha_k \leq \alpha$ through a union bound. This only covers the specified audited deployment blocks. It does not certify policy π_{k+1} trained after that calibration, nor reuse of calibration labels to choose among arbitrary new policies. Uniform function sets or properly justified sequential inference are needed for broader claims.

Weighted calibration is another comparator when labels have a fixed conditional mechanism and reference/on-policy density ratios are available [34]. In text, action-product ratios can be extremely variable. Report effective sample size $(\sum_i w_i)^2 / \sum_i w_i^2$, overlap, and clipping bias. Holding the completer fixed is necessary for the success-probability target; updating the completer changes that target.

Potential contribution. Measure and repair post-optimization coverage loss, showing fewer confidently wrong selected trajectories than static sets, ensembles, and hard-negative training. Main risks are label cost, feedback loops, catastrophic shared verifier mistakes, and moving targets. Treat an adversarial curriculum as empirical adaptation, not automatic uniform robustness.

17. Theoretical characterization: established lemmas and open targets

The propositions below are elementary supporting results derived for these notes. They are not claimed as original publishable theorems. They identify the assumptions a stronger result must supply.

Proposition 1 (Rectangular monotone aggregation). *For a finite trajectory and the independent box $\prod_t [L_t, U_t]$, if G is coordinatewise nondecreasing, then $\inf_{\mathbf{r} \in \prod_t [L_t, U_t]} G(\mathbf{r}) = G(\mathbf{L})$.*

Proof. Every feasible vector is coordinatewise at least $\mathbf{L}$, and $\mathbf{L}$ is feasible. Monotonicity gives both inequalities. For a coupled set $\mathbf{L}$ may not be feasible; its value then remains a conservative lower bound, not an exact robust score. $\square$

Proposition 2 (A simultaneous tube protects a selected candidate). *Suppose $R_j^* \in [L_j, U_j]$ for every member of a finite candidate pool. Let $\hat{j} \in \arg \max_j L_j$, and let $j^* \in \arg \max_j R_j^*$. Then*

$$R_{\hat{j}}^* \geq L_{\hat{j}}, \quad R_{j^*}^* - R_{\hat{j}}^* \leq U_{j^*} - L_{j^*}.$$

Proof. $R_{\hat{j}}^* \geq L_{\hat{j}} \geq L_{j^*}$ and $R_{j^*}^* \leq U_{j^*}$. Subtract. The simultaneous containment event is needed before selection. For half-widths bounded by ϵ , regret is at most 2ϵ . If these are process-aggregate rewards, the bound concerns that aggregate and not automatically final-answer correctness. $\square$

17.1. Why marginal coverage alone fails

Construct a distribution where a fraction α of candidates are wrong, receive score one, and receive an interval that excludes the true reward zero. All other candidates are correct, score 0.8, and are covered. Random-candidate coverage is exactly $1 - \alpha$. Best-of- N chooses an uncovered wrong candidate whenever one occurs, with probability $1 - (1 - \alpha)^N$. At $\alpha = 0.05$, $N = 100$, this is about 0.994. Ties are resolved within wrong candidates. This is a constructed counterexample, not a measured PRM failure. It illustrates why selection-aware theory is necessary even before RL.

Proposition 3 (Robust improvement against a reference). *If the true reward function belongs to a fixed set $\mathcal{U}$, then for every policy*

$$\Delta_{\mathcal{U}}(\pi) := \inf_{r \in \mathcal{U}} \{J_r(\pi) - J_r(\pi_0)\} \leq J_{r^*}(\pi) - J_{r^*}(\pi_0).$$

Therefore $\Delta_{\mathcal{U}}(\pi) > 0$ certifies improvement for that target reward.

Proof. Evaluate the infimum at r^* . The same containment event works for adaptively chosen policies if it is a uniform function-containment event on their domain. $\square$

This reference-centered objective is preferable to comparing two separate pessimistic values: improvement in $\inf_r J_r(\pi)$ does not alone prove improvement in the true return. Approximate occupancy estimates, optimization error, and uncertain labels each require an explicit error allowance.

17.2. A shift-aware bound with transparent assumptions

Suppose a bounded trajectory utility satisfies $R^* = \hat{R} + e$ and $\mathbb{E}_{P_0} e^2 \leq \epsilon^2$, with $P_\pi \ll P_0$. For $w = dP_\pi/dP_0$, Cauchy–Schwarz gives

$$\left| \mathbb{E}_{P_\pi} R^* - \mathbb{E}_{P_\pi} \hat{R} \right| \leq \epsilon \sqrt{\mathbb{E}_{P_0} w^2} = \epsilon \sqrt{1 + \chi^2(P_\pi \| P_0)}.$$

For improvement relative to P_0 , the error difference is bounded by $\epsilon \sqrt{\chi^2(P_\pi \| P_0)}$, using $w - 1$. This is a standard change-of-measure calculation related to correlated-proxy regularization [18]; it fails when relevant support is absent or the ratio is unbounded. Replacing whole trajectories by step occupancy needs consistent normalization and an additive reward objective.

17.3. Open theoretical targets

1. **Geometry versus conservatism:** characterize regret or safe improvement for boxes versus shared ellipsoids at equal joint coverage, including correlated misspecification and variable-length reasoning. A dual-norm identity alone is insufficient novelty.
2. **Coverage under specified optimization:** prove selected-trajectory coverage for a feasible protocol beyond naive NT union bounds, explicitly comparing with JOMI and problem-block calibration. Specify the target and budget.
3. **Necessity and lower bounds:** construct families where any narrow set calibrated only on a reference policy misses a reachable adversarial mode; state the needed overlap or complexity assumptions for informative protection.
4. **Repair-aware process uncertainty:** characterize a set over validity transitions and show how robust outcome estimation differs from weakest-step scoring, comparing with RSP. This is more ambitious and needs reliable labels.

A convergence theorem for a neural min–max method should not be asserted from convex-head or strongly monotone toy assumptions. Prove the restricted model first, and state the empirical extension separately.

18. An experiment plan that can support a paper

18.1. Hypotheses and staged scope

- H1:** Static pointwise coverage degrades among PRM-selected candidates as search pressure increases.
- H2:** A calibrated tube reduces false high-reward selections better than equal-cost pessimism, label improvement, or reduced search.
- H3:** Coupling improves the coverage–sharpness–accuracy tradeoff when errors share structure, but can fail under unmodeled common bias.
- H4:** Static calibration degrades under policy movement; independently audited updates recover useful coverage in specified regimes.

Begin with frozen-pool inference and a synthetic setting with known truth. Proceed to RL only if uncertainty is informative and an overoptimization baseline can be reproduced. A pure empirical paper can stop at inference-time claims; it should not imply mitigation of RL hacking without RL experiments.

18.2. Data and split protocol

Use PRM800K or appropriate process training data for heads and adapters, with training, validation, calibration, and test splits separated by problem. All paraphrases, injected errors, sampled solutions, and prefixes of a problem belong to one split. Use ProcessBench, PRMBench, and Socratic-PRMBench as independent diagnostic benchmarks; do not train a calibrator on their final evaluation labels.

For answer evaluation, generate trajectories on held-out MATH-family problems, with GSM-style tasks as a second difficulty regime. Small AIME sets are a stress test with wide uncertainty, not the sole source of statistical claims. If a published baseline calibrates on MATH500, make that data use explicit and evaluate on a distinct held-out set. Check upstream training contamination where possible, and qualify residual uncertainty.

Use at least two generator sizes or families and at least two PRM families, such as Qwen2.5-Math-PRM and Skywork-o1 PRMs as documented in the cited work. These are reproducibility candidates, not claims about the latest model releases. Verify model revisions and inference interfaces before running experiments.

18.3. Baselines and ablations

Comparator	Confound it addresses
Nominal single PRM; calibrated center	Is the benefit better point predictions?
Mean, minimum, and variance-penalized ensembles	Standard reward pessimism
AdvPO-style frozen-head confidence score	Existing confidence-region optimization
Quantile PRM; conditional-OT calibration	Existing prefix uncertainty modeling
Caution novelty penalty	Single-model OOD conservatism
PURE-style minimum; sum/mean/product controls	Credit assignment versus uncertainty
Reduced N , early stopping, and KL sweep	Less optimization versus a better reward
Length correction and hard-negative retraining	Bias correction and improved labels
Box versus coupled tube; calibrated versus tuned radius	Geometry and calibration
Pointwise, block, union-bound, selection-conditional variants	What survives selection

Comparator	Confound it addresses
Random widths; constant widths; oracle error widths	Whether width identifies errors
Reliable outcome-only RLVR; hybrid rewards	Whether a PRM improves a reliable task signal

A full factorial is expensive. Predefine a core set containing nominal, calibrated-center, ensemble pessimism, AdvPO, one PRM calibration method, Caution, and the proposed tube; use targeted secondary ablations. Match center quality, training labels, total generation/scoring cost, and where relevant measured KL. Radius must be calibrated on calibration data or tuned on validation, never on the test set. Report those variants separately.

18.4. Optimization stress tests

For frozen candidates, sweep $N \in \{1, 4, 16, 64, 256\}$, subject to a common token budget, scoring nested candidate pools. Report both equal- N and equal-compute comparisons. Test step splitting/merging, repetition, neutral padding, incorrect numeric transformations, unsupported premises, prompt mismatch, suffix attacks, and delimiter manipulation. Validate whether each edit preserves or changes correctness, and keep an untouched transfer family.

For RL, freeze a reward model and tube during each comparison run. Use equal initial policies, prompts, token caps, training compute, and KL coefficients; add a matched-observed-KL analysis because penalties can change policy movement. The primary controlled comparison assigns the same trajectory aggregation to nominal and robust rewards. Compare process-only and hybrid-outcome supervision separately. A maximum-empty-response score is a parser defect to fix and report, not a gain to accept. Save checkpoints and raw rewards before normalization.

Attack the deployed robust score, including its uncertainty head. A defense that works only against attacks optimized for $\hat{r}$ may simply move the exploit to $\hat{r} - b$. Black-box search and white-box token attacks are separate budgets; report access assumptions, calls, failed attempts, and transfer results.

18.5. Metrics and statistical reporting

The primary metric is held-out answer correctness under increasing pressure, supplemented by independent process validity on a stratified audit subset. Use answer checkers with canonicalization and spot checks; do not call a larger LLM an exact oracle. Score trajectories with correct answers but flawed processes as a separate category.

Report step, trajectory, candidate-pool, and selected-trajectory coverage; interval width and singleton/abstention rates; false acceptance of invalid steps; overestimation error among top-scoring candidates; reward-accuracy curves; length, repetition, vacuity, KL, and compute. For bounded aligned utilities, also report lower-bound violations $(L - R^*)_+$ and their tail severity. Do not subtract an uncalibrated free scalar reward from a binary accuracy metric and interpret the result as a meaningful numerical “hacking gap.”

Use paired candidate pools, multiple independent head/policy seeds, and problem-level bootstrap confidence intervals. A starting plan is at least three seeds for expensive RL and five for

lightweight methods; justify additional runs from variance and a pilot power analysis. Resampling individual steps as if independent understates uncertainty. Report all prespecified pressure levels, including unsuccessful conditions. Specify checkpoints by validation or fixed training budget, never retrospective test accuracy.

18.6. Synthetic experiments with known ground truth

Build finite reasoning trees with known validity labels and optional repair transitions. Expose features containing a true validity signal and spurious style dimensions. Vary reference-policy coverage, shared versus independent reward error, trajectory length, repair rate, and optimization budget. Compare nominal search/RL, boxes, ellipsoids, bias sets, and oracle bounds. This identifies when the set really contains truth and when apparent conservatism only shrinks optimization. Synthetic results validate assumptions, not realistic LLM claims.

19. Recommended route and publication decision gates

Route	Minimum convincing result	Main limitation
Empirical	Selected-path coverage failures plus a useful, equal-cost repair on multiple PRMs and pressure regimes	Prediction sets may be too wide; no RL claim without RL evidence
Theoretical	A process-specific coverage/regret or impossibility result beyond standard confidence-set lemmas	Strong feature/support assumptions may limit practical implications
Combined	A restricted theorem, informative real-PRM tubes, and adaptation-aware stress tests that match the theorem’s mechanism	Highest label and training cost

Recommended starting point: Approach A as a diagnostic and calibrated baseline, Approach B as the structured candidate method, and synthetic tests to separate shared uncertainty from pointwise errors. Add a misspecification term from Approach C only when common-mode failures are observed. Approach E is the RL extension; Approach D is a later alternative, not a prerequisite.

1. **Target gate:** a label and reward definition must be coherent. If success-probability bounds are the only informative ones, narrow the claim to that target instead of calling them correctness bounds.
2. **Information gate:** uncertainty must predict overestimation or useful abstention better than constant/random widths. Otherwise improve the representation or report why this uncertainty signal fails.
3. **Mechanism gate:** show gains at fixed aggregation and comparable compute/optimization pressure. If reduced search explains the effect, do not claim a stronger reward estimator.

4. **Prior-work gate:** compare explicitly with AdvPO, PRM quantile/OT calibration, ensemble pessimism, Caution, and selection-aware inference. Update the novelty search before drafting a submission.
5. **Robustness gate:** test adaptive attacks and held-out families. If calibration fails under RL, retain an honest inference-only result or study the failure as the main contribution.

The immediate deliverable is a research design, not an experiment result or a guarantee of publication. A well-supported negative finding about tube coverage may also be valuable, provided it is broad, controlled, and explains a mechanism that existing uncertainty methods miss.

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